Everpure_FBEXA_file_32hosts_10-DN-DNOS-1.1.0-4845253c3899_30-S500R2-blades-MDN-4.8.3.exa-4845253c

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Testbed Summary:

FlashBlade//EXA cluster summary:

Metadata Node Purity//FBEXA version: 4.8.3.exa-4845253c
Data Node Purity//DN (DNOS) version: 1.1.0-4845253c3899
Total healthy FlashBlade//EXA blades: **30**
Total healthy DataFabric Modules (DFMs): **120**

All available data nodes were used:
- mlperf node group name: **10-datanodes**
- number of datanodes in mlperf node group: **10**

Worker host initiators:
- Physical worker host initiator count: **32**

Datanode RAID configuration:

Software RAID striped-mirror layout for the data volume.
RAID 1+0, implemented as mdadm RAID0 over RAID1

Data RAID layout:
- 16 x KIOXIA KCM7DRJE3T84 NVMe SSDs are used as data drives.
- The 16 data SSDs are organized as 8 mdadm RAID1 mirror arrays.
- The RAID1 mirror arrays are /dev/md120 through /dev/md127.
- Each RAID1 array contains 2 whole NVMe devices and uses mdadm metadata version 1.2.
- The top-level array is /dev/md119.
- /dev/md119 is an mdadm RAID0 array striped across the 8 RAID1 mirror arrays.
- The RAID0 array uses a 32 KiB mdadm chunk size.
- An XFS filesystem is created directly on /dev/md119.
- /dev/md119 is mounted at /export/1320565f-8067-4d6d-9a80-0ae958632725.
- The resulting layout is RAID0 over RAID1, commonly described as striped mirrors.

FlashBlade//EXA is Pure's extreme-scale file storage platform for very large AI and HPC environments, built around a **disaggregated architecture** that separates metadata services from data services so each can scale independently.

Technically, it has three main building blocks:

- **Metadata Nodes (MN):** Pure-supplied FlashBlade-based metadata core, storing filesystem metadata on DFMs and connecting through XFMs.
- **Data Nodes (DN):** customer/partner-sourced off-the-shelf Linux servers with SSDs/NVMe that store user data and connect directly to the customer network.
- **XFMs:** extended fabric modules that provide network connectivity for the metadata domain.

From a protocol and I/O standpoint, EXA behaves like a **parallel NFS system**:

- Clients talk to the **metadata tier** using **NFSv4.1/pNFS over TCP** to get file layout and location information.
- Clients then access the **data nodes directly** using **NFSv3 over RDMA** for the high-speed data path.
- The metadata node decides which data node owns a file, and at GA a file maps to a particular data node rather than being striped across nodes in the classic parallel-file-system sense.

The networking model is deliberately modern and hyperscale-oriented:

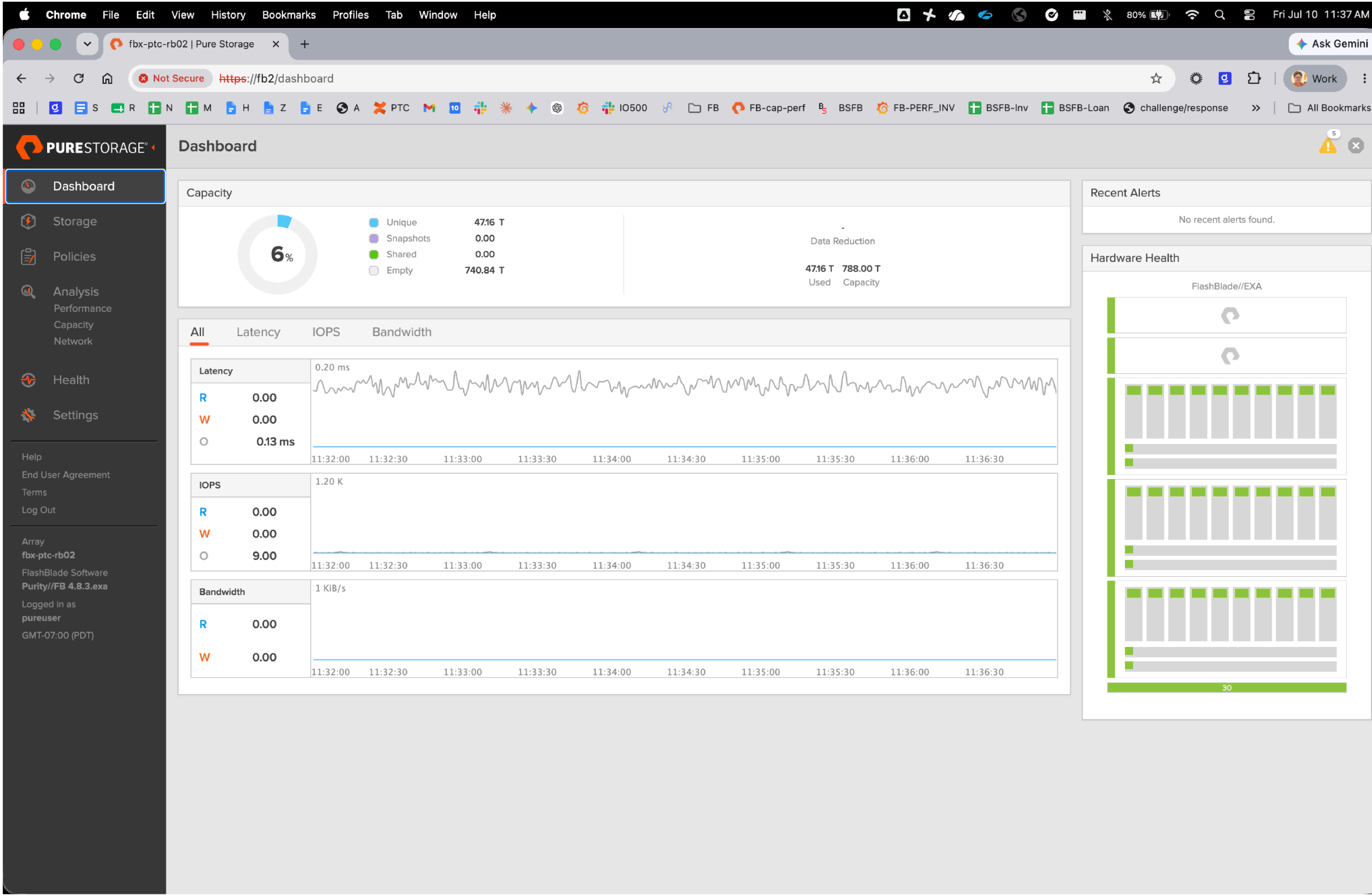
- It expects a **Layer 3 leaf/spine fabric** using **eBGP** and **ECMP** rather than traditional L2/MLAG-centric storage networking.
- Data movement uses **RoCEv2/RDMA** for low-latency, high-throughput access to data nodes.

At the storage layer:

- Data nodes run **XFS** on top of **Linux MD-RAID**, with configurable **RAID50** or **RAID10** depending on whether the goal is better efficiency or maximum performance.
- EXA is optimized for throughput and metadata speed rather than efficiency features, so **data reduction (dedupe/compression) is not a design goal** in the core architecture.

In practical terms, FlashBlade//EXA is best thought of as a **scale-out, disaggregated, pNFS-based file platform** designed to feed massive GPU clusters with very high throughput while avoiding metadata bottlenecks and preserving operational simplicity compared with legacy parallel file systems.

FlashBlade//EXA main GUI Dashboard


Chrome File Edit View History Bookmarks Profiles Tab Window Help
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Shared 0.00
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Data Reduction
47.16 T Used
788.00 T Capacity
Recent Alerts
No recent alerts found.
Hardware Health
FlashBlade//EXA
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W 0.00
O 0.13 ms
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IOPS
R 0.00
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1 KiB/s
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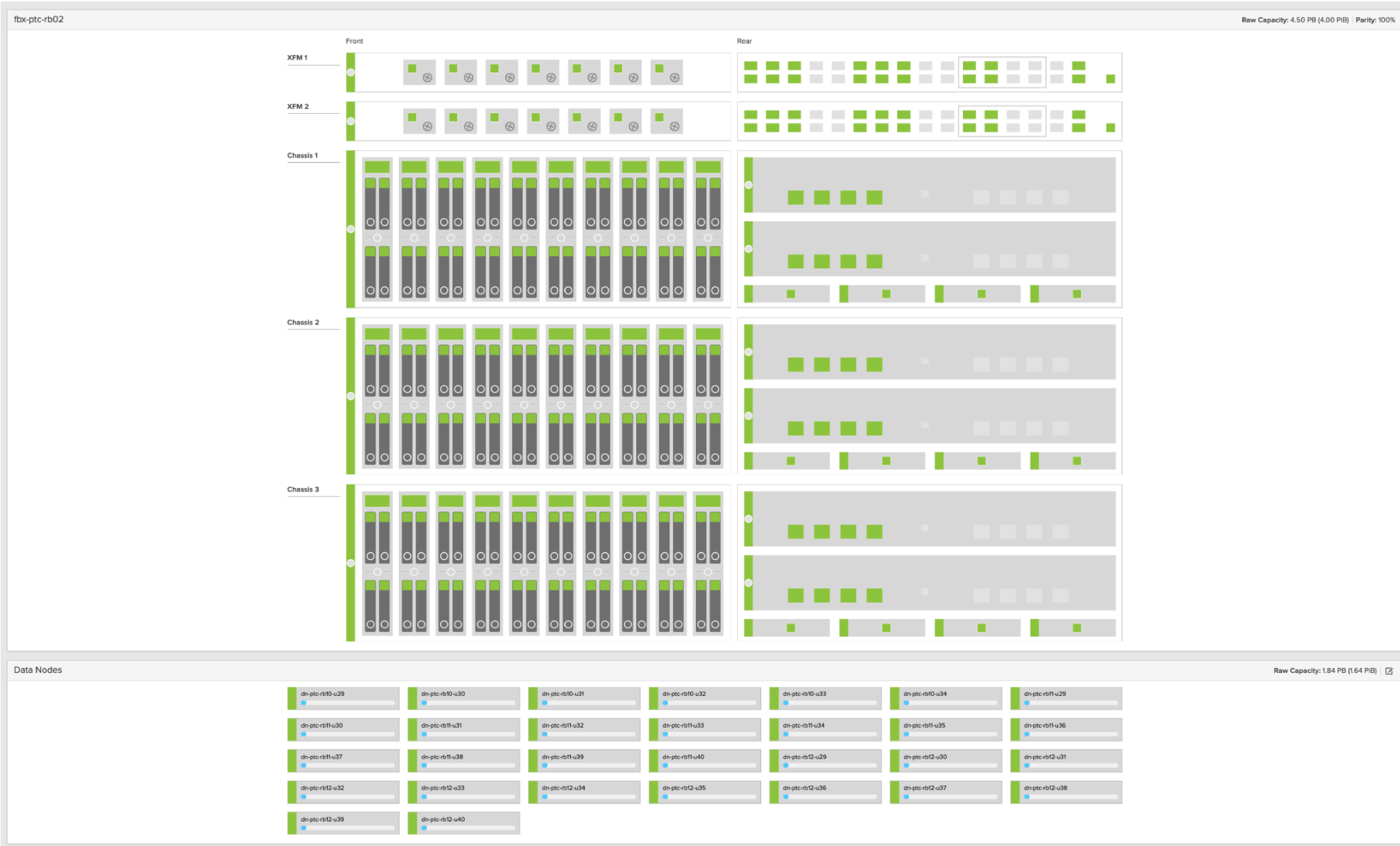
====

Hardware component health view



XFM-8400 is Pure's **second-generation external fabric module** for FlashBlade multi-chassis systems: effectively a **pair of 1U Ethernet switches** that interconnect multiple FlashBlade chassis into one larger system and provide host/uplink networking for that multi-chassis domain.

Technically, the XFM-8400 is:

- a **32-port 400GbE switch platform** running a Purity//FB image.
- used in **multi-chassis FlashBlade** deployments, where an XFM pair can scale a system to **up to 10 chassis**.

From a connectivity perspective:

- **Ports 21–28** are used for **host/customer network attachment**.
- The lower-numbered ports are used for **downlinks to FlashBlade chassis/FIOMs** in multi-chassis topologies.
- It supports **100G uplinks** initially, plus **200G** and **400G uplinks** on supported Purity//FB releases and optics/transceivers.

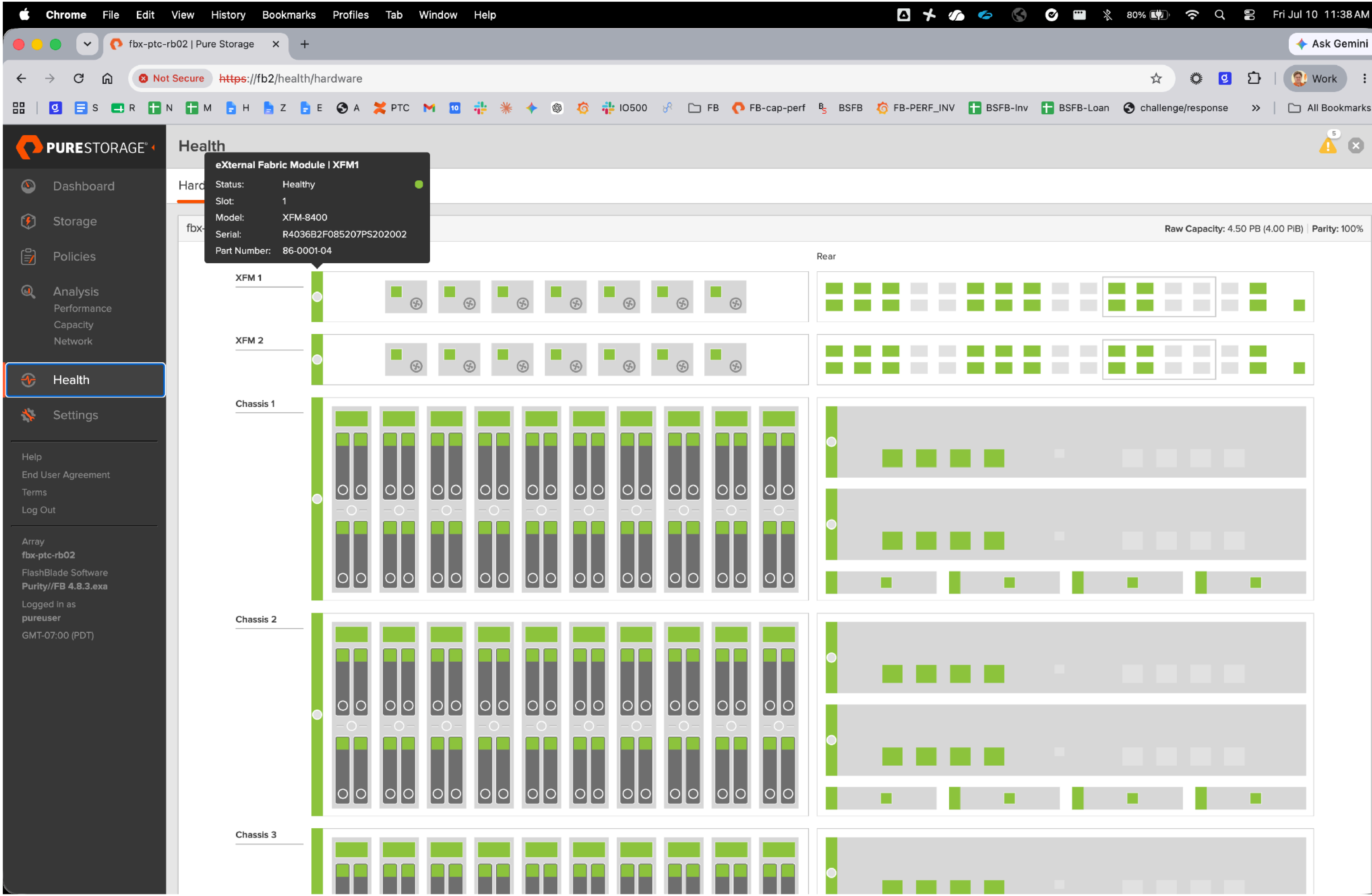
Architecturally, its main role is to remove the chassis-to-chassis scaling limits of earlier designs:

- it is **required for multi-chassis systems**, especially larger topologies beyond 5 chassis.
- it supports dedicated **ISL links** between the XFM pair, with **100G ISL** on XFM-8400 vs **40G ISL** on the older EFM-3200e.
- it enables higher aggregate bandwidth between chassis and to the network, including newer **4x100G breakout downlink** modes on XFM-8400/8400R2 for high-throughput systems.

Operationally:

- XFM-8400 is part of the **HA/failover fabric**, where one XFM can stay active while the peer is rebooted or serviced.
- It can be introduced via **non-disruptive upgrade** from EFM-3200e on supported systems and releases.
- It has hardware-specific compatibility gates tied to **minimum Purity//FB versions**; for example XFMv2.0 support starts in the 4.x Purity//FB line, while newer XFM-8400 R2 hardware requires later releases.

In short, **XFM-8400 is the high-speed multi-chassis Ethernet fabric switch layer for FlashBlade**, designed to aggregate chassis connectivity, provide uplinks to the customer network, and enable larger-scale FlashBlade topologies with 100/200/400GbE-era bandwidth and HA behavior.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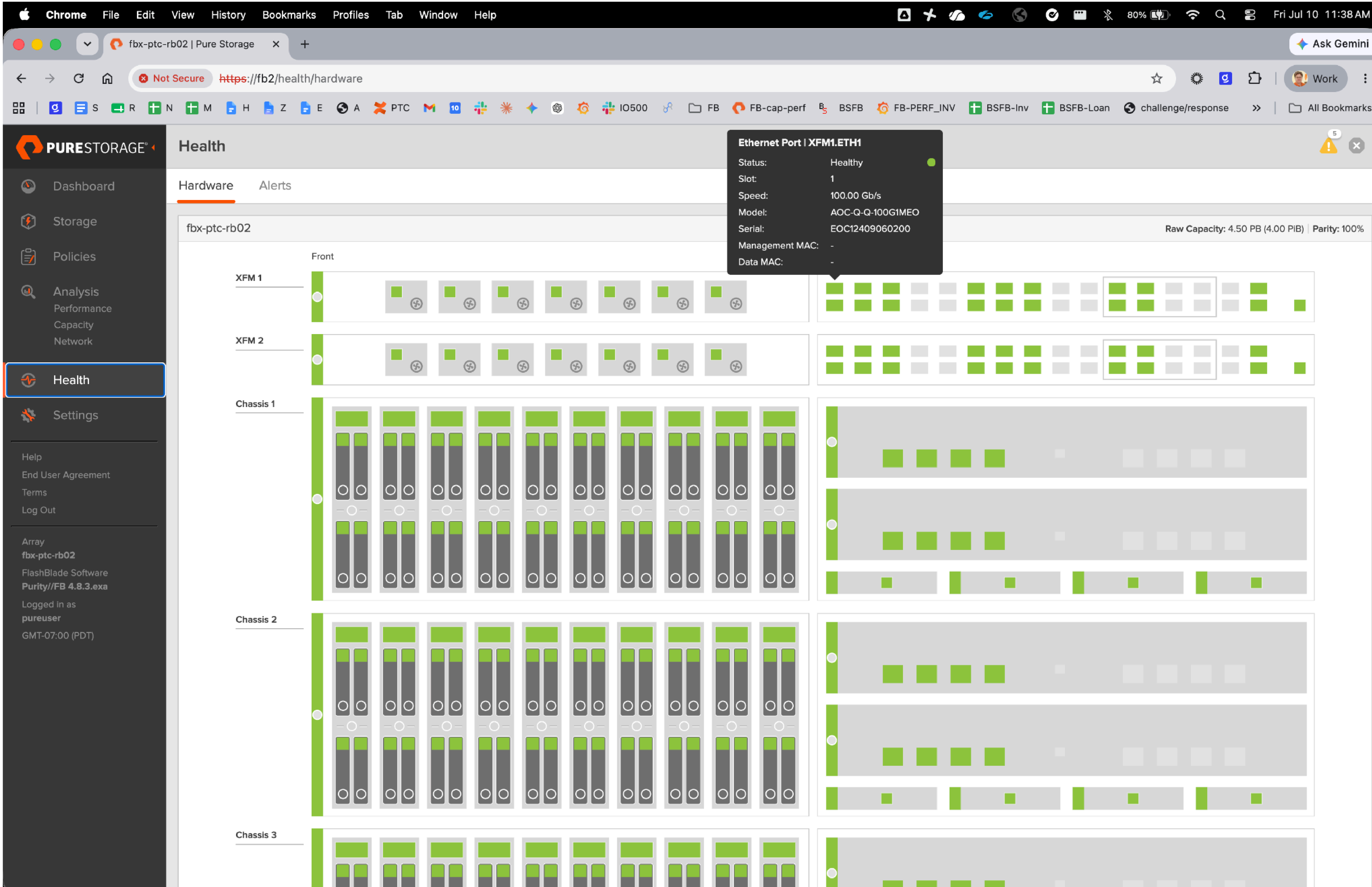
Hard
fbx-
eXternal Fabric Module | XFM1
Status: Healthy
Slot: 1
Model: XFM-8400
Serial: R4036B2F085207PS202002
Part Number: 86-0001-04
Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Rear
XFM 1
XFM 2
Chassis 1
Chassis 2
Chassis 3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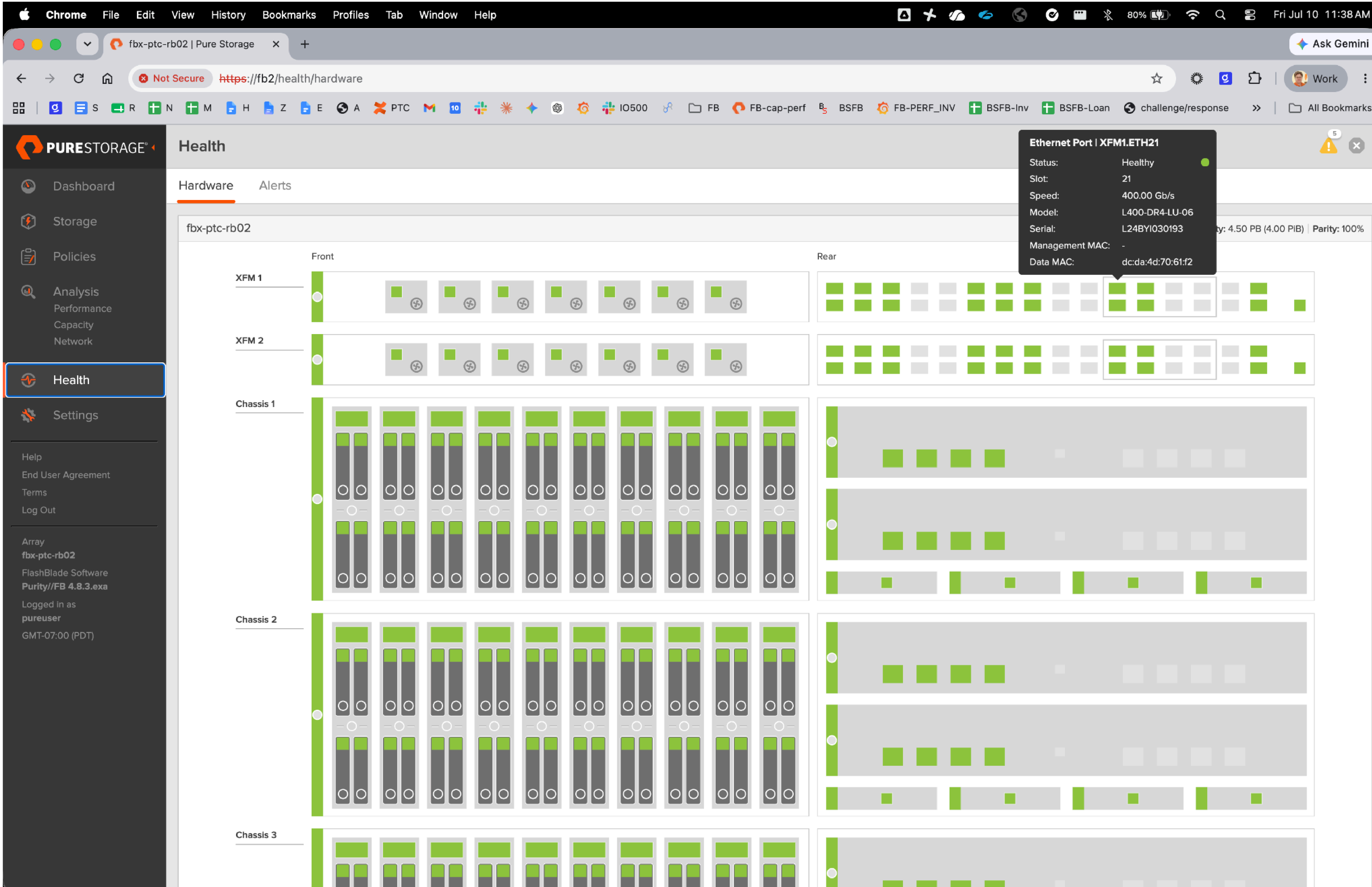
Alerts
fbx-ptc-rb02
Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Ethernet Port | XFM1.ETH1
Status: Healthy
Slot: 1
Speed: 100.00 Gb/s
Model: AOC-Q-Q-100G1MEO
Serial: EOC12409060200
Management MAC: -
Data MAC: -
Front
XFM 1
XFM 2
Chassis 1
Chassis 2
Chassis 3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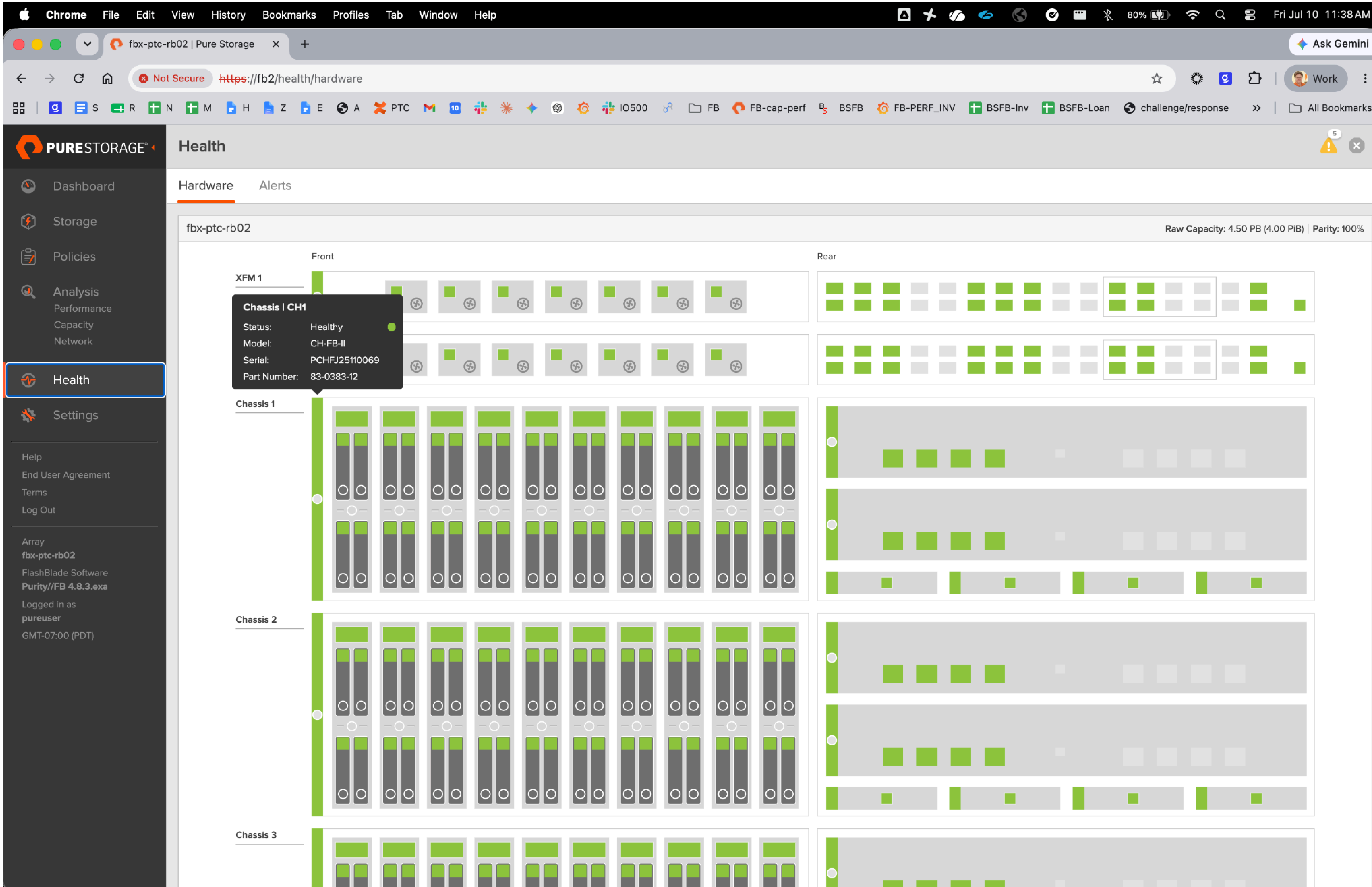
y: 4.50 PB (4.00 PiB) | Parity: 100%
Ethernet Port | XFM1.ETH21
Status: Healthy
Slot: 21
Speed: 400.00 Gb/s
Model: L400-DR4-LU-06
Serial: L24BYI030193
Management MAC: -
Data MAC: dc:da:4d:70:61:f2
Front
Rear
XFM 1
XFM 2
Chassis 1
Chassis 2
Chassis 3

CH-FB-II is the **5RU second-generation FlashBlade chassis** used for the Legend/Meteor hardware family—the physical enclosure that hosts modern FlashBlade blade-based systems rather than being a storage node by itself.

- **5 rack units (5RU)** in height
- **10 blade slots** used to house FlashBlade//EXA blades
- **2 Fabric Module (FIOM) bays**
- **4 power supply bays**
    - **4 power supplies**
    - **2+2 PSU redundancy**
    - **4.8 kW max rated chassis power**.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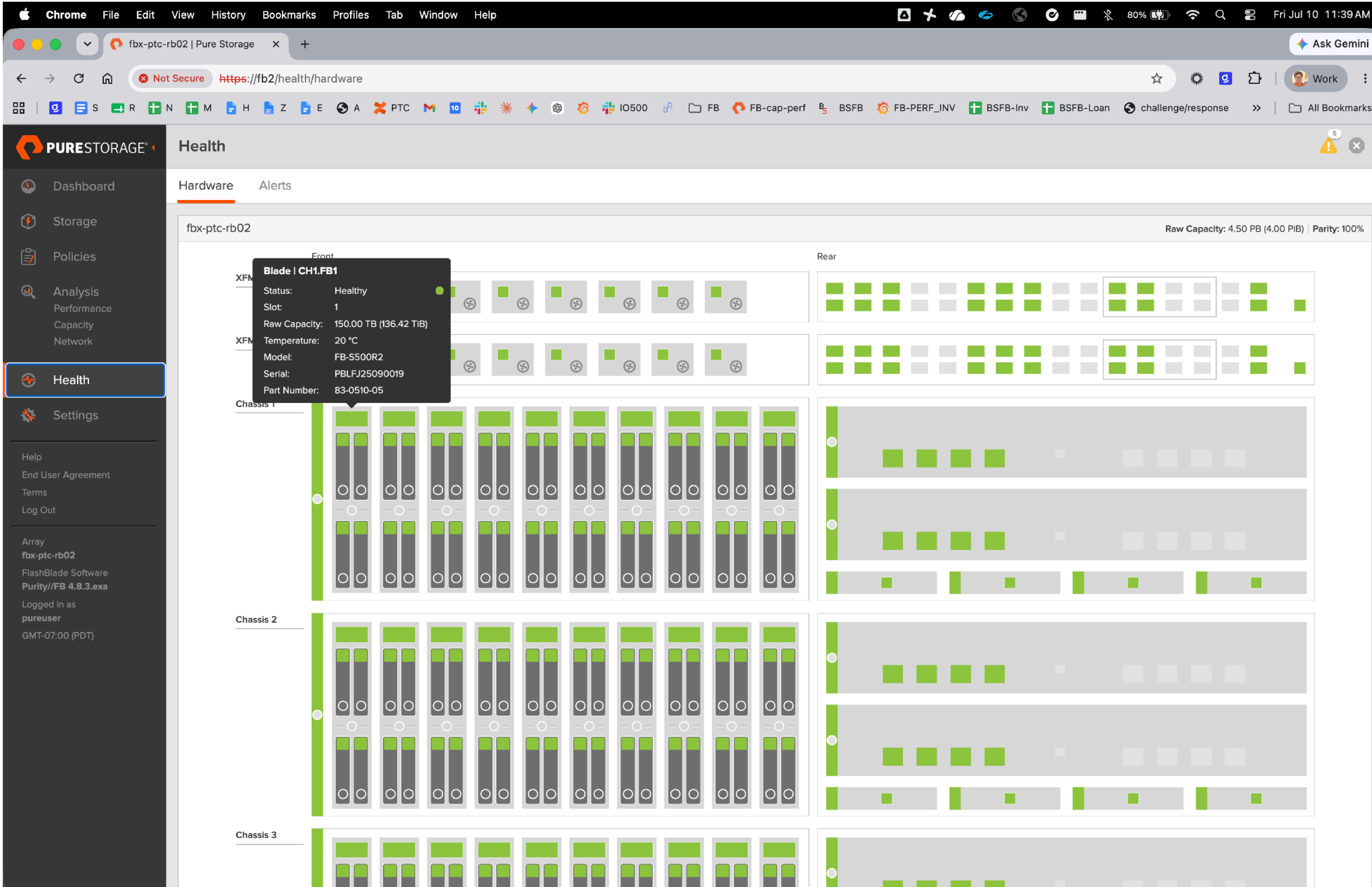
fbx-ptc-rb02
Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Front
Rear
XFM 1
Chassis | CH1
Status: Healthy
Model: CH-FB-II
Serial: PCHFJ25110069
Part Number: 83-0383-12
Chassis 1
Chassis 2
Chassis 3

FB-S500R2 is the **second-generation high-performance FlashBlade//EXA blade** for the CH-FB-II chassis

- **Gen4 PCIe** connectivity to the storage media path.
- **2 x 100GbE CX6-DX data-plane interfaces** over Gen4 PCIe.
- Support for **up to 4 DFMs per blade**, with at least one required.

Within the chassis/system architecture:

- An FB//EXA chassis can hold **up to 10 FB-S500R2 blades**.
- Each S500R2 blade connects internally to each FIOM at **100GbE**.
- R2 blades are **hot-swappable** and fit into the existing FlashBlade//EXA chassis for **non-disruptive in-place upgrades**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GMT-07:00 (PDT)
Hardware
Alerts
fbx-ptc-rb02
Raw Capacity: 4.50 PB (4.00 PiB)
Parity: 100%
Front
Rear
Blade | CH1.FB1
Status: Healthy
Slot: 1
Raw Capacity: 150.00 TB (136.42 TiB)
Temperature: 20 °C
Model: FB-S500R2
Serial: PBLFJ25090019
Part Number: 83-0510-05
Chassis 2
Chassis 3

DirectFlash Module (DFM)

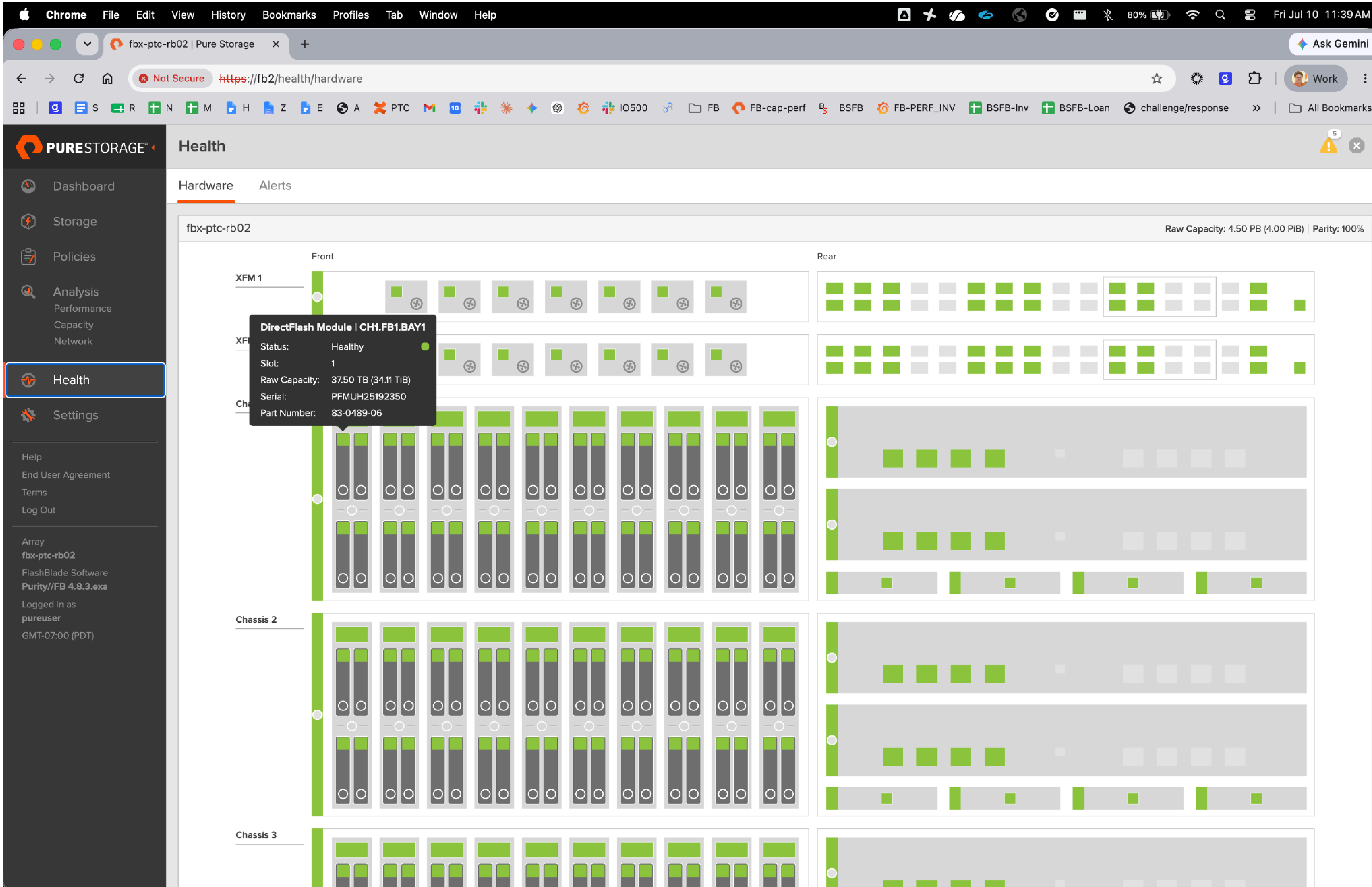
83-0489-06 is a **37.5 TB raw DirectFlash Module (DFM)** for FlashBlade, providing **34.11 TiB usable nominal capacity per module** in the system sizing/spec context.

- **Flash generation: N58R**
- **NAND type: QLC**

In system architecture terms, it is the **data-bearing storage module that plugs into FlashBlade//EXA blade bays**.

Practically, think of it as:

- a **single FlashBlade media module**
- contributing **37.5 TB raw** each, so a **10-blade / 4-DFM-per-blade** build with this module yields **40 DFMs total** at the chassis level.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Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Front
Rear
XFM 1
DirectFlash Module | CH1.FB1.BAY1
Status: Healthy
Slot: 1
Raw Capacity: 37.50 TB (34.11 TiB)
Serial: PFMUH25192350
Part Number: 83-0489-06
Chassis 2
Chassis 3

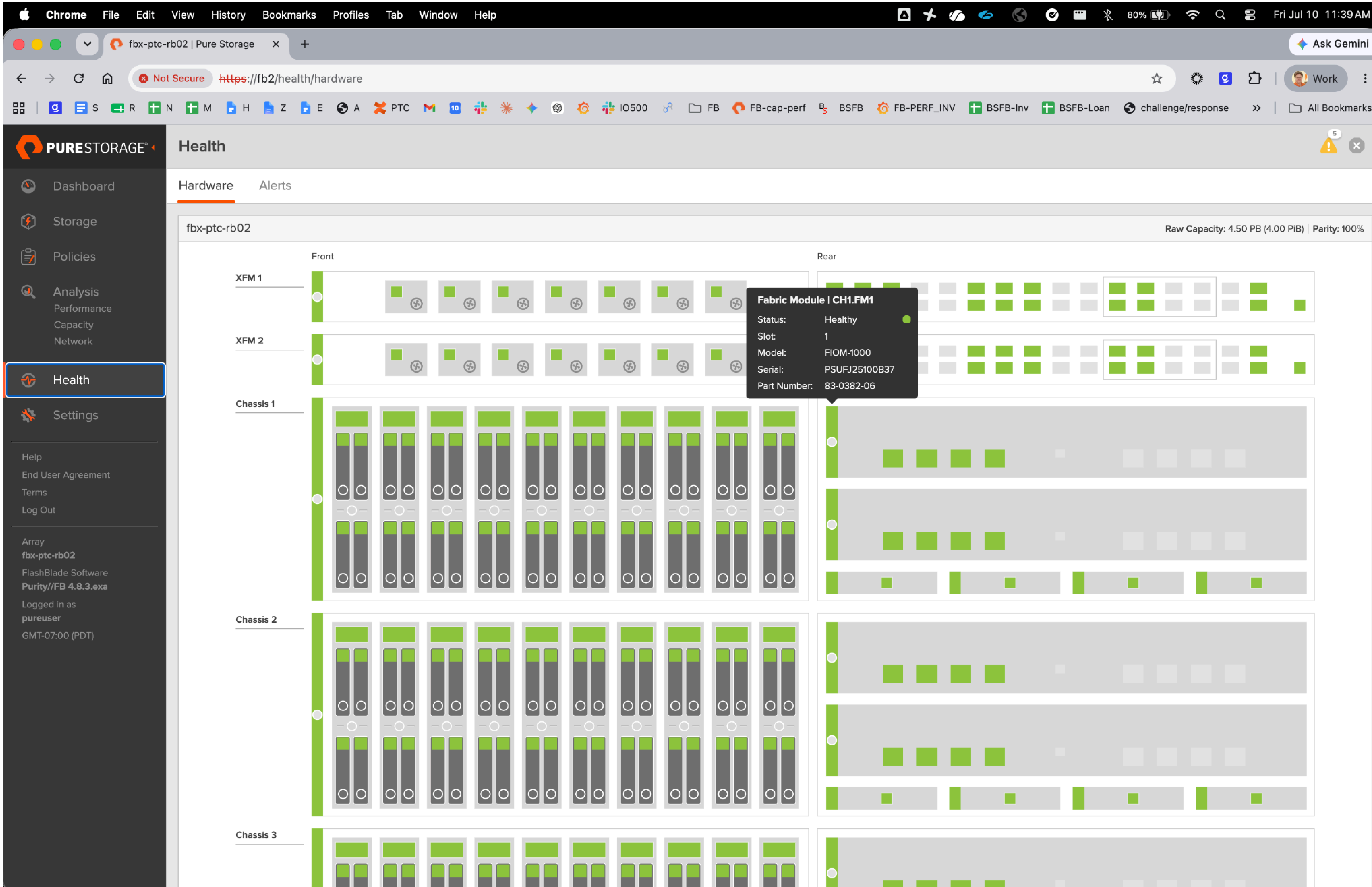
Fabric Module (FIOM)

Technically, it serves as the **per-chassis control/fabric module** rather than a storage blade:

- There are **two FIOMs per chassis** for redundancy/HA.
- It supports **up to 10 blades per chassis**, **8 uplink ports**, and **2 ISL links**
- It is a powered board with its own sensors, GPIO, EEPROM, PMBus, temperature, fan, voltage, USB-device, and upgrade interfaces.

Operationally, the FIOM is an **active compute/control element**:

- The FIOM is responsible for **supervision, local switching/fabric control, and chassis management-plane functions** inside the FlashBlade//EXA MDN system.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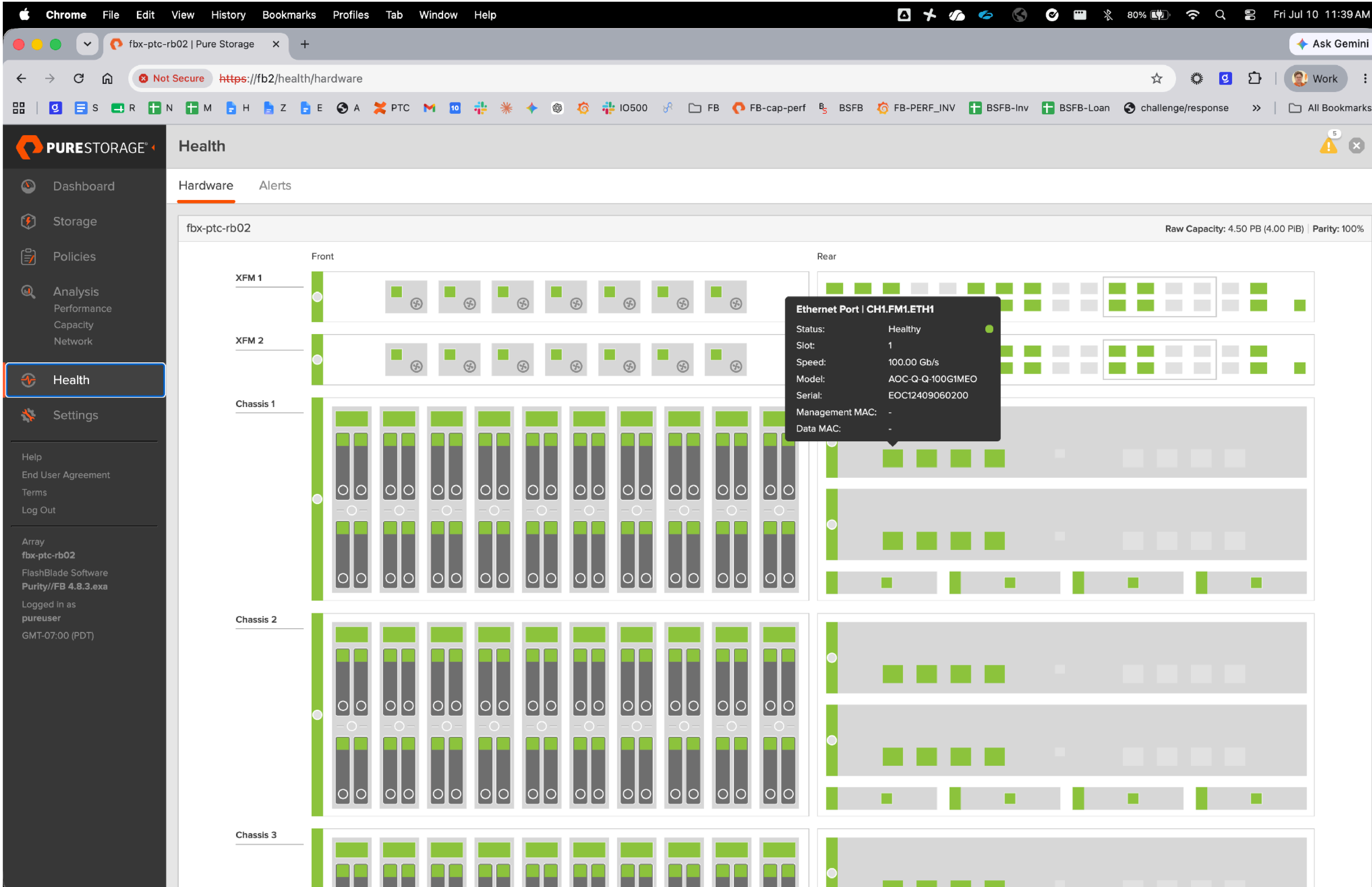
Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Front
Rear
XFM 1
XFM 2
Chassis 1
Chassis 2
Chassis 3
Fabric Module | CH1.FM1
Status: Healthy
Slot: 1
Model: FIOM-1000
Serial: PSUFJ25100B37
Part Number: 83-0382-06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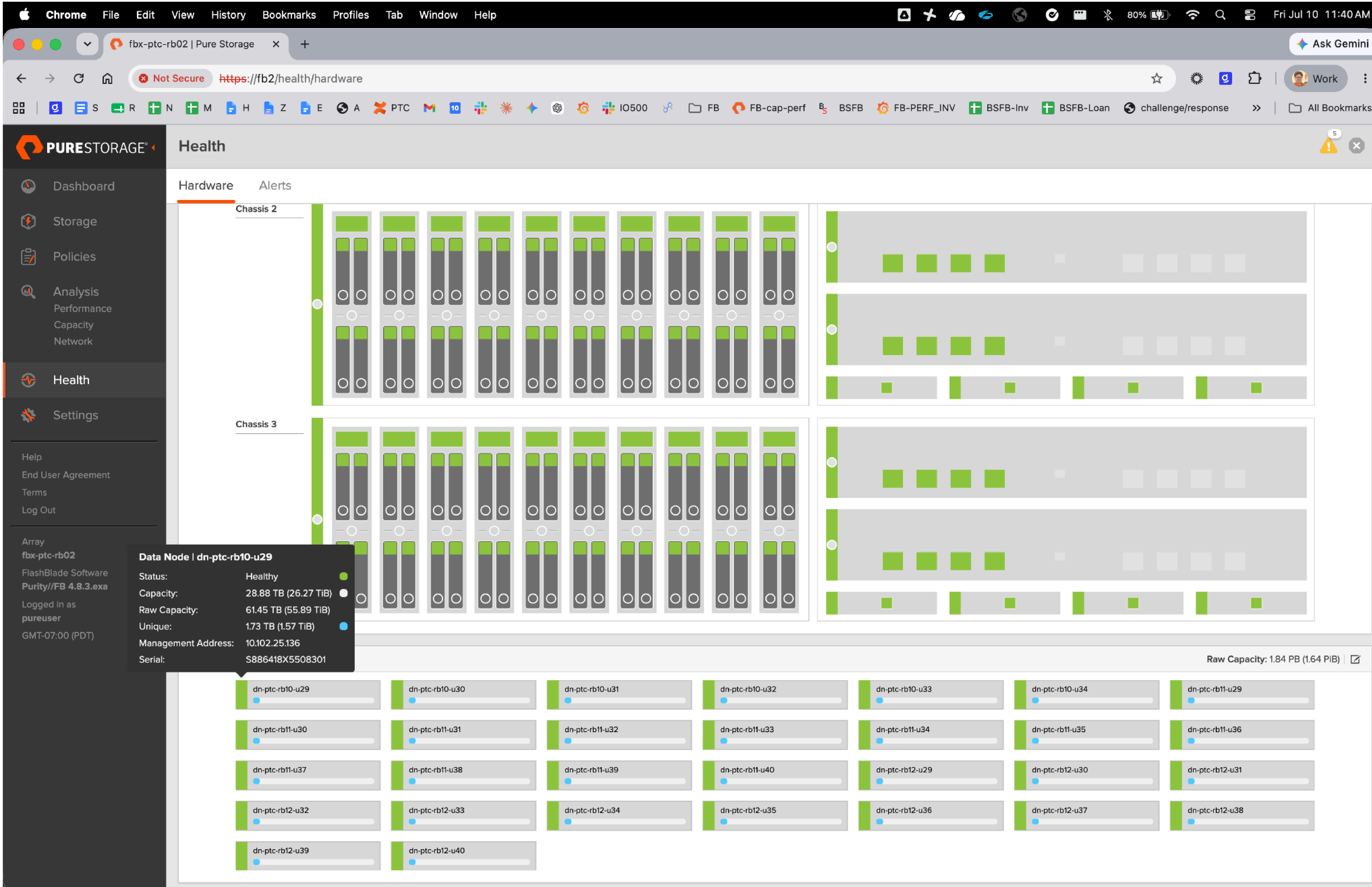
Raw Capacity: 4.50 PB (4.00 PiB) | Parity: 100%
Front
Rear
XFM 1
XFM 2
Chassis 1
Chassis 2
Chassis 3
Ethernet Port | CH1.FM1.ETH1
Status: Healthy
Slot: 1
Speed: 100.00 Gb/s
Model: AOC-Q-Q-100G1MEO
Serial: EOC12409060200
Management MAC: -
Data MAC: -

30 Data nodes were available to this cluster (However, for this test only 10 were used)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FlashBlade Software
Purity//FB 4.8.3.exa
Logged in as
pureuser
GMT-07:00 (PDT)
Hardware
Alerts
Chassis 2
Chassis 3
Data Node | dn-ptc-rb10-u29
Status: Healthy
Capacity: 28.88 TB (26.27 TiB)
Raw Capacity: 61.45 TB (55.89 TiB)
Unique: 1.73 TB (1.57 TiB)
Management Address: 10.102.25.136
Serial: S886418X5508301
Raw Capacity: 1.84 PB (1.64 PiB)
dn-ptc-rb10-u29
dn-ptc-rb10-u30
dn-ptc-rb10-u31
dn-ptc-rb10-u32
dn-ptc-rb10-u33
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Work

PURESTORAGE

Settings

- Dashboard
- Storage
- Policies
- Analysis
  - Performance
  - Capacity
  - Network
- Health
- Settings

Help
End User Agreement
Terms
Log Out

Array
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FlashBlade Software
Purity//FB 4.8.3.exa
Logged in as
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GMT-07:00 (PDT)

> Data Nodes

Search Settings

SYSTEM
- Alerts and Notifications
- Array Details
- Attached Policies
- Monitoring
- NTP Servers
- Support

HARDWARE
- Blades
- Data Nodes
- Drives
- Hardware

NETWORK
- Connectors
- Diagnostics
- DNS
- Object Store Virtual Hosts
- Subnets & Interfaces

SECURITY
- API Clients
- Audit Trail
- Certificates
- Directory Service
- Kerberos Keytabs
- Public Keys
- Session Log
- Single Sign-On
- Users

Data Node Groups

1-1 of 1

Name ▲	ID
all-datanodes	54b4c825-bb57-9345-3769-342706cee052

Data Nodes

General Space 1-10 of 30

Name ▲	Management Address	Data Addresses	Serial Number	Chassis	Resiliency Group	Status	Details
dn-ptc-rb10-u29	10.102.25.136	192.168.120.136	S886418X5508301	C1260MO11A30191	-	healthy	-
dn-ptc-rb10-u30	10.102.25.135	192.168.120.135	S886418X5508297	C1260MO11A30323	-	healthy	-
dn-ptc-rb10-u31	10.102.25.134	192.168.120.134	S886418X5508300	C1260MO11A30322	-	healthy	-
dn-ptc-rb10-u32	10.102.25.133	192.168.120.133	S886418X5508347	C1260MO11A30138	-	healthy	-
dn-ptc-rb10-u33	10.102.25.132	192.168.120.132	S886418X5508416	C1260MO11A30097	-	healthy	-
dn-ptc-rb10-u34	10.102.25.131	192.168.120.131	S886418X5508387	C1260MO11A30213	-	healthy	-
dn-ptc-rb11-u29	10.102.25.124	192.168.120.124	S886418X5508395	C1260MO11A30310	-	healthy	-
dn-ptc-rb11-u30	10.102.25.123	192.168.120.123	S886418X5508393	C1260MO11A30314	-	healthy	-
dn-ptc-rb11-u31	10.102.25.122	192.168.120.122	S886418X5508362	C1260MO11A30247	-	healthy	-
dn-ptc-rb11-u32	10.102.25.121	192.168.120.121	S886418X5508373	C1260MO11A30239	-	healthy	-

Resiliency Groups

No resiliency groups found.

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fbx-ptc-rb02 | Pure Storage

Not Secure ~~https~~://fb2/settings/subnets

S R N M H Z E A PTC IO500 FB FB-cap-perf BSFB FB-PERF_INV BSFB-Inv BSFB-Loan challenge/response All Bookmarks

PURESTORAGE

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Array
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Logged in as
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> Subnets & Interfaces

Search Settings

SYSTEM
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SECURITY
- API Clients
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- Public Keys
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- Users

Subnets

1-2 of 2

Name ▲	Enabled	Prefix	VLAN	Gateway	MTU	LAG	Services
							Any
data	True	192.168.2.0/24	0		9000		data
mgmt-v4	True	10.102.24.0/22	0	10.102.24.1	1500		management, support

Interfaces

1-10 of 30

Name ▲	Enabled	Server	Subnet	Address	VLAN	Mask	Gateway	MTU	Service
									Data
data_192_168_2_101	True	_array_server	data	192.168.2.101	0	255.255.255.0		9000	data
data_192_168_2_102	True	_array_server	data	192.168.2.102	0	255.255.255.0		9000	data
data_192_168_2_103	True	_array_server	data	192.168.2.103	0	255.255.255.0		9000	data
data_192_168_2_104	True	_array_server	data	192.168.2.104	0	255.255.255.0		9000	data
data_192_168_2_105	True	_array_server	data	192.168.2.105	0	255.255.255.0		9000	data
data_192_168_2_106	True	_array_server	data	192.168.2.106	0	255.255.255.0		9000	data
data_192_168_2_107	True	_array_server	data	192.168.2.107	0	255.255.255.0		9000	data
data_192_168_2_108	True	_array_server	data	192.168.2.108	0	255.255.255.0		9000	data
data_192_168_2_109	True	_array_server	data	192.168.2.109	0	255.255.255.0		9000	data
data_192_168_2_110	True	_array_server	data	192.168.2.110	0	255.255.255.0		9000	data

A single FlashBlade//EXA file system was created.

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80% Fri Jul 10 11:41 AM

fbx-ptc-rb02 | Pure Storage

Ask Gemini

Not Secure https://fb2/storage/array

S R N M H Z E A PTC IO500 FB FB-cap-perf BSFB FB-PERF_INV BSFB-Inv BSFB-Loan challenge/response All Bookmarks

Work

PURESTORAGE

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Array
fbx-ptc-rb02
FlashBlade Software
Purity//FB 4.8.3.exa
Logged in as
pureuser
GMT-07:00 (PDT)

Storage

Array | Realms | Servers | File Systems | Object Store

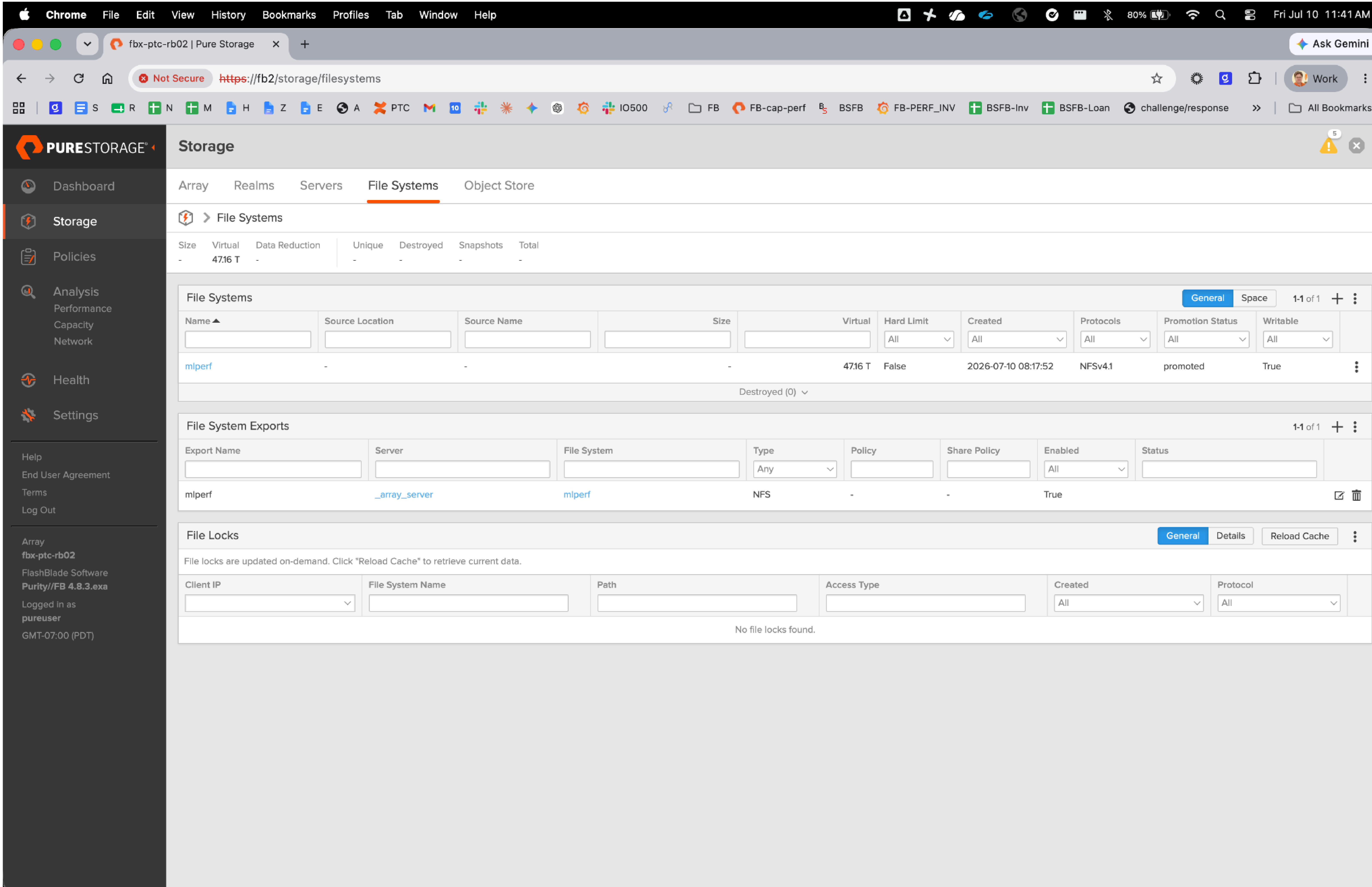
Array

Virtual	Data Reduction	Unique	Destroyed	Snapshots	Shared	Total	Capacity
47.16 T	-	47.16 T	-	-	-	47.16 T	788.00 T

fbx-ptc-rb02 ID f0df8140-4b9c-4e09-a6af-7c43bed4ceea

Realms	Servers	File Systems	Network Access Policies	NFS Export Policies	SSH Certificate Authority Policies	TLS Policies	Object Store Accounts	Object Store Users	Object Store Roles
0	1	1	1	0	0	1	0	0	0

Object Store Access Policies	Object Store S3 Export Policies	Object Store Buckets	Object Store Objects	Object Store Virtual Hosts	DNS	Network Interfaces	Certificates	Certificate Groups
13	1	0	-	1	1	33	1	1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Logged in as
pureuser
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Array Realms Servers File Systems Object Store
File Systems
Size - Virtual 47.16 T Data Reduction - Unique - Destroyed - Snapshots - Total -
File Systems
General Space 1-1 of 1
Name Source Location Source Name Size Virtual Hard Limit Created Protocols Promotion Status Writable
mlperf - - - 47.16 T False 2026-07-10 08:17:52 NFSv4.1 promoted True
Destroyed (0)
File System Exports
1-1 of 1
Export Name Server File System Type Policy Share Policy Enabled Status
mlperf _array_server mlperf NFS - - True
File Locks
General Details Reload Cache
File locks are updated on-demand. Click "Reload Cache" to retrieve current data.
Client IP File System Name Path Access Type Created Protocol
No file locks found.

Chrome File Edit View History Bookmarks Profiles Tab Window Help 80% Fri Jul 10 11:41 AM

fbx-ptc-rb02 | Pure Storage

Not Secure https://fb2/storage/filesystems/mlperf

S R N M H Z E A PTC IO500 FB FB-cap-perf BSFB FB-PERF_INV BSFB-Inv BSFB-Loan challenge/response All Bookmarks

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Storage

Array | Realms | Servers | File Systems | Object Store

File Systems > mlperf

Size	Virtual	Data Reduction	Unique	Snapshots	Total
-	47.16 T	-	-	-	-

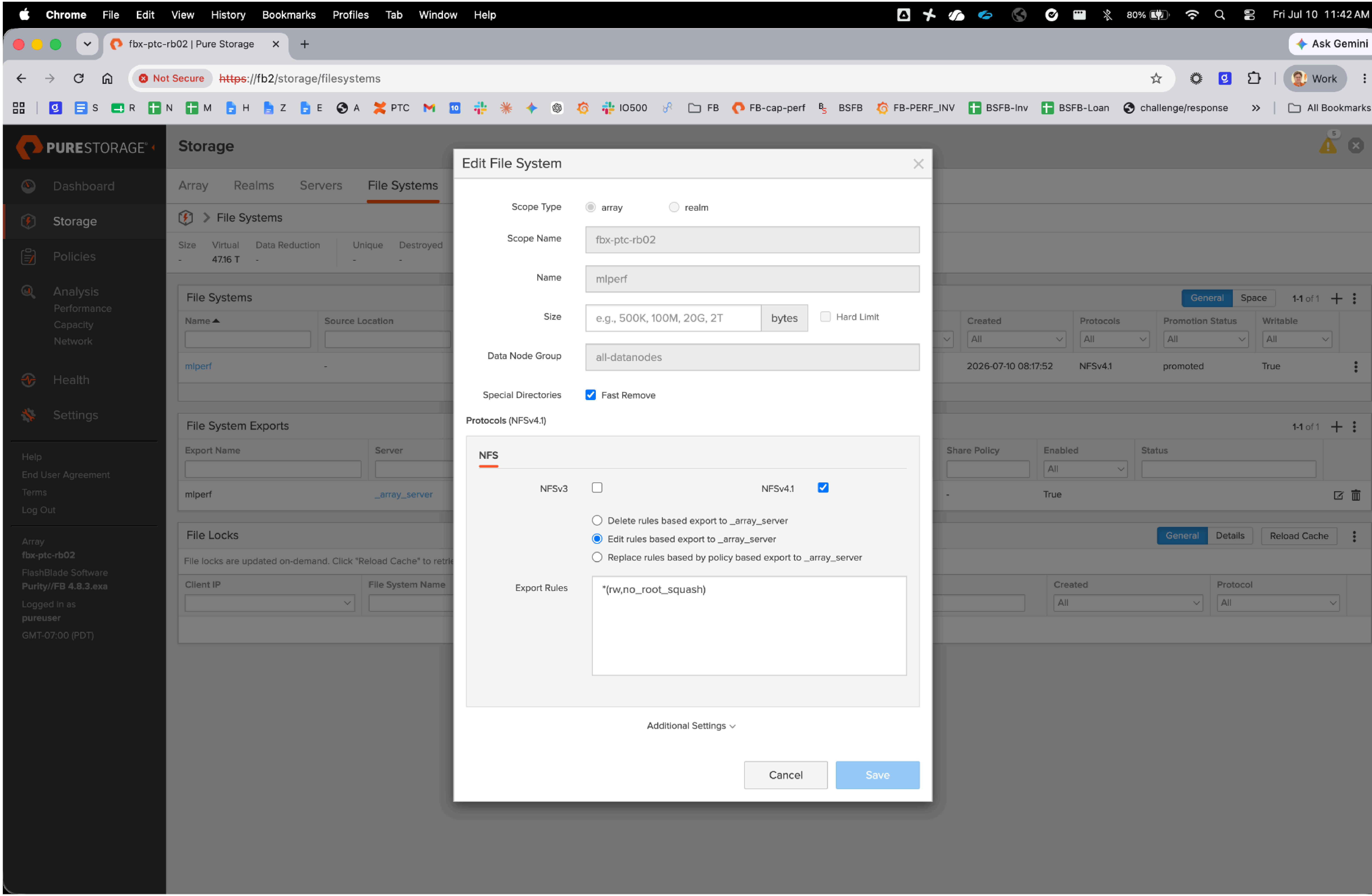
File System Exports

1-1 of 1

Export Name	Server	Type	Policy	Share Policy	Enabled	Status
		Any			All	
mlperf	_array_server	NFS	-	-	True	

Details

Created	2026-07-10 8:17:52	Quotas »		Protocols »	
Group Ownership	creator	**Default Group Quota**	-	**Enabled Protocols**	NFSv4.1
Hard Limit	False	**Default User Quota**	-	**Access Control Style**	shared
Promotion Status	promoted			**Safeguard ACLs**	True
Source	-				
Writable	True				
Node Group	all-datanodes				



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https://fb2/storage/filesystems
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Realms
Servers
File Systems
File Systems
Size
Virtual
Data Reduction
Unique
Destroyed
47.16 T
File Systems
Name
Source Location
mlperf
File System Exports
Export Name
Server
mlperf
_array_server
File Locks
File locks are updated on-demand. Click "Reload Cache" to retri
Client IP
File System Name
General
Space
1-1 of 1
Created
Protocols
Promotion Status
Writable
2026-07-10 08:17:52
NFSv4.1
promoted
True
Share Policy
Enabled
Status
True
General
Details
Reload Cache
Protocol
Edit File System
Scope Type
array
realm
Scope Name
fbx-ptc-rb02
Name
mlperf
Size
e.g., 500K, 100M, 20G, 2T
bytes
Hard Limit
Data Node Group
all-datanodes
Special Directories
Fast Remove
Protocols (NFSv4.1)
NFS
NFSv3
NFSv4.1
Delete rules based export to _array_server
Edit rules based export to _array_server
Replace rules based by policy based export to _array_server
Export Rules
*(rw,no_root_squash)
Additional Settings
Cancel
Save

FlashBlade//EXA Metadata Node (MDN)

The Metadata Node system was a **3-chassis** multi-chassis configuration with 2 eXternal Fabric Modules (XFMs).
Each XFM had 4 x 400Gbps Uplink ports.
Each chassis is connected to each XFM with 4x100Gb uplink ports.
Each chassis was equipped with 10 x S500R2 FlashBlade//EXA blades.
Each blade was equipped with 4 × 37.5TB N58R DirectFlash Modules (DFMs).

Operating System: Purity//FBEXA version: 4.8.3.exa-4845253c

Metadata Node system details:

- 2 x eXternal Fabric Modules (XFMs) - Model: XFM-8400 - Part Number: 86-0001-04
- **3** x FlashBlade Chassis - Model: CH-FB-II - Part Number: 83-0383-12
- 10 x FlashBlade S500R2 blades per chassis - Model: FB-S500R2 - Part Number: 83-0510-05
- 4 x DirectFlash Modules (DFMs) per blade - Raw Capacity: 37.50 TB (34.11 TiB) - Part Number: 83-0489-06

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Tackling Myths Around AI Data and FlashBlade//EXA

Inside Pure Storage's FlashBlade//EXA: Scaling AI Without Bottlenecks - Six Five In The Booth

This is an SSD?! - PureStorage FlashBlade Tour

Technical Deep Dive on FlashBlade//S | Everpure

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XFM power specifications

For **XFM-8400, part number 86-0001-04**, the power specs are:

- **Input voltage:** 200–240 VAC
- **Frequency:** 50/60 Hz
- **Power factor correction:** ≥95%
- **Inrush current:** 40 A per power supply unit
- **Power supplies:** 2 high-efficiency **80 Plus Platinum** AC PSUs

Peak for a single XFM-8400 module:

- **AC draw:** 600 W
- **AC draw:** 630 VA
- **Current @ 208 V:** 2.9 A
- **Current @ 240 V:** 2.5 A
- **Heat:** 2,046 BTU/hr

Typical for a single XFM-8400 module:

- **AC power:** 314 W

- **AC power:** 330 VA
- **Current @ 208 V:** 1.5 A
- **Current @ 240 V:** 1.3 A
- **Heat:** 1,071 BTU/hr

Optics overhead noted in the same spec:

- **Downlinks:** 14 W per chassis
- **Uplinks:** 10 W per link

Also, the module includes **two IEC C13/C14 (10A 250V) power cords**.

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Fully-populated FlashBlade//EXA MDN chassis power specifications

- FlashBlade Chassis - Model: CH-FB-II - Part Number: 83-0383-12
- 10 x FlashBlade S500R2 blades per chassis - Model: FB-S500R2 - Part Number: 83-0510-05
- 4 x DirectFlash Modules (DFMs) per blade - Raw Capacity: 37.50 TB (34.11 TiB) - Part Number: 83-0489-06

System Configuration

Item	**Specification**
Chassis Model	CH-FB-II
Chassis Part Number	83-0383-12
Blade Configuration	10 × FB-S500R2 blades
Blade Part Number	83-0510-05
DFM Configuration	4 × DFMs per blade
DFM Capacity	37.50 TB raw (34.11 TiB) each
DFM Part Number	83-0489-06
Fully Configured System	FB-S500R2-1500TB-37DFMc

Power and Thermal Specifications

Parameter	**Specification**
Nominal Power	2,771 W
Typical Power	2,595 W
Peak Power	3,181 W
Peak Apparent Power	3,340 VA
Typical Heat Dissipation	9,291 BTU/hr
Peak Heat Dissipation	11,390 BTU/hr

Parameter	Specification
Rack Height	5 RU

Input Current

Condition	208 V	240 V
Typical Current	13.1 A	11.4 A
Peak Current	16.1 A	13.9 A

Power Delivery

Parameter	Specification
Power Supplies per Chassis	4
Redundancy	2+2
Power Supply Rating	2,400 W each
Chassis Max Rated Power	4,800 W
Input Voltage	200–240 VAC nominal
Power Cord Type	IEC C19/C20

Notes

The exact matched configured system row for this build is the 1-chassis S500R2 1500 TB configuration, listed at 2,771 W nominal power.

Published S500 chassis data states that the chassis power figures are independent of storage capacity and DFM count.

R2 blades are documented as consuming slightly more power than R1 blades, while PSU and accessory support remain unchanged.

Datanode Information

Supermicro ASG-1115S-NE316R servers

- CPU = [single-socket AMD EPYC 9355P processor with 32 physical cores (64 logical CPUs via SMT) on a 64-bit x86 architecture]
- MEMORY = (192 GB of DDR5 ECC memory).
- DATA NETWORK ADAPTERS = (2 x NVIDIA ConnectX-7 EN (MT2910) single-port 400 GbE QSFP112 Ethernet adapters installed, operating over PCIe Gen5 with secure firmware and InfiniBand/VPI functionality disabled.)
- SSDs
    - Each data node has 16 x KIOXIA KCM7DRJE3T84 3.84 TB enterprise NVMe SSDs installed.
    - Each data node had 61.45 TB of raw capacity and 28.88 TB of usable capacity.
- Operating System
    - The FlashBlade//EXA Data Node Operating System (Purity//DN) was loaded onto each data node.
    - Purity//DN (DNOS) version: 1.1.0-4845253c3899
- RAID layout
    - RAID 10–style layout: (Inner layer: consists of RAID 1 mirrors of NVMe drive pairs) (Outer layer: RAID 0 striped across 8 mirrored devices) (Filesystem: XFS on top of the striped mirror set) The system uses an mdadm RAID0-over-RAID1 (striped mirrors) configuration with an XFS filesystem on top; this results in an inherent 50% capacity efficiency due to mirroring, making approximately 866 TB of usable capacity from 1843.5 TB of raw capacity mathematically consistent once RAID metadata, XFS filesystem structures, and alignment overhead are included.

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Single-Server Power Specifications

System: Supermicro ASG-1115S-NE316R data node
Configuration: 1 x AMD EPYC 9355P, 192 GB DDR5 ECC, 2 x NVIDIA ConnectX-7 400 GbE adapters, 16 x KIOXIA KCM7DRJE3T84 3.84 TB E3.S NVMe SSDs

Specification	Value
Power supply configuration	2 x 1600 W redundant 1+1 Titanium-level PSUs
PSU efficiency rating	Titanium level, up to 96%
Recommended input voltage	200–240 Vac, 50/60 Hz
Low-line input support	100–127 Vac, but PSU output derates to 1000 W
High-line PSU output rating	1600 W per redundant PSU at 200–240 Vac
Estimated active DC load	~894 W DC
Estimated active AC input power	~931 W AC
Estimated high-load DC load	~1,014 W DC
Estimated high-load AC input power	~1,079 W AC
Recommended facility power allocation	**1.2 kW AC per server**
Nameplate allocation option	1.6 kW output capacity per server at 200–240 Vac
Estimated active current at 208 Vac	~4.7 A
Estimated active current at 240 Vac	~4.1 A
Estimated high-load current at 208 Vac	~5.5 A
Estimated high-load current at 240 Vac	~4.7 A
Recommended allocation current at 208 Vac	~6.1 A, based on 1.2 kW and 0.95 PF
Recommended allocation current at 240 Vac	~5.3 A, based on 1.2 kW and 0.95 PF
Estimated active heat output	~3,177 BTU/hr
Estimated high-load heat output	~3,680 BTU/hr
Recommended cooling allowance	~4,094 BTU/hr per server, based on 1.2 kW allocation

Component Power Basis

Component	Quantity	Power Basis	Subtotal

AMD EPYC 9355P CPU	1	280 W default TDP	280 W
KIOXIA KCM7DRJE3T84 NVMe SSD	16	24 W active typical	384 W
NVIDIA ConnectX-7 400 GbE adapter	2	24.9 W typical each with passive cables	49.8 W
DDR5 ECC memory	192 GB	Estimated	~60 W
Motherboard, BMC, fans, backplane, boot media, platform overhead	1 system	Estimated	~120 W active / ~200 W high-load allowance

Calculated active load:
280 W + 384 W + 49.8 W + 60 W + 120 W = 893.8 W DC

Estimated active AC input:
893.8 W ÷ 0.96 = ~931 W AC

High-load planning load:
300 W CPU allowance + 384 W SSDs + 49.8 W NICs + 80 W memory allowance + 200 W platform/fan allowance = ~1,014 W DC

Estimated high-load AC input:
1,014 W ÷ 0.94 = ~1,079 W AC

The Supermicro ASG-1115S-NE316R chassis supports a single AMD EPYC 9004/9005 processor up to 300 W TDP, 16 hot-swap E3.S NVMe bays, up to 8 heavy-duty 4 cm fans, and 2 x 1600 W redundant Titanium-level power supplies. Supermicro lists the PSUs as 1000 W at 100–127 Vac and 1600 W at 200–240 Vac, so this configuration should be specified for **200–240 Vac input**. (Supermicro)

The AMD EPYC 9355P is a 32-core / 64-thread processor with 280 W default TDP and 240–300 W configurable TDP. (AMD) The KIOXIA KCM7DRJE3T84 is the 3,840 GB CM7-R E3.S SED model, with 24 W typical active power and 5 W typical ready power listed for the 3.84 TB capacity class. (KIOXIA America, Inc.) The NVIDIA ConnectX-7 MCX715105AS-WEAT is a single-port QSFP112 adapter supporting 400/200/100/50/25/10 GbE, with 24.9 W typical power listed for passive cables in PCIe Gen5 x16 operation. (networking-docs.nvidia.com)

Specification-sheet note: Power estimates exclude additional QSFP112 optical transceiver power. Add the transceiver/module power separately when optical modules are installed.

Client Host initiators

physical worker host initiator count: 32

32 x Supermicro SYS-621C-TN12R servers
- CPU = dual-socket Intel Xeon Silver 4516Y+ platform with 48 physical cores (96 logical CPUs via SMT) on a 64-bit x86 architecture
- MEMORY = 1024 GB of DDR5 ECC memory
  - for the retinanet training tests, the memory was limited by GRUB to 188 GiB of memory
    - GRUB_CMDLINE_LINUX_DEFAULT="quiet splash mem=198752M"

```
# free -h
               total        used        free      shared  buff/cache   available
Mem:           188Gi        10Gi        13Gi       6.0Mi       166Gi       178Gi
Swap:             0B          0B          0B
```

- DATA NETWORK ADAPTERS
  - 2 x NVIDIA ConnectX-7 EN (MT2910) single-port 400 GbE QSFP112 Ethernet adapters installed, operating over PCIe Gen5 with secure firmware and InfiniBand/VPI functionality disabled.
  - RDMA-enabled
- SSD = 1 x Micron 5400-series MTFDDAK240TGA 240 GB enterprise SATA SSD (2.5-inch, SATA 6 Gb/s) with full SMART support and 100% remaining endurance, used as the system disk.
- Operating System = 24.04.4 LTS (Noble Numbat)
- Linux kernel: 6.18.26-061826-generic

Example of NFSv4.1 mount to a MDN data vip from each client host initiator:

192.168.2.103:/mlperf on /mnt/mlperf type **nfs**4
(rw,relatime,vers=4.1,rsize=524288,wsize=524288,namlen=255,hard,fatal_neterrors=none,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=192.168.10.103,local_lock=none,addr=192.168.2.103)

Networking Stack

- pNFS with RDMA-enabled 400Gbps NICs
- This includes all firmware and kernel-level drivers supporting RoCE (RDMA over Converged Ethernet) for ultra-low latency.
- Each host initiator and each data node has 2 x NVIDIA (Mellanox) ConnectX-7 EN (MT2910) single-port 400 GbE QSFP112 Ethernet adapters installed, operating over PCIe Gen5 with secure firmware and InfiniBand/VPI functionality disabled.
- BGP (Border Gateway Protocol) networking was used.
- Native OS-provided NFS client, drivers, and tools were used.
- RDMA over Converged Ethernet (RoCEv2) was used across the entire fabric.

Single-Server Power Specifications

System: Supermicro SYS-621C-TN12R
Configuration: 2 x Intel Xeon Silver 4516Y+ CPUs, 1024 GB DDR5 ECC, 2 x NVIDIA ConnectX-7 EN 400 GbE adapters, 1 x Micron 5400-series MTFDDAK240TGA 240 GB SATA SSD
Operating system: Ubuntu 24.04.4 LTS, Linux kernel 6.18.26-061826-generic

Specification	Value
Power supply configuration	2 x 1200 W redundant 1+1 Titanium-level hot-plug PSUs
PSU efficiency rating	Titanium level, up to 96%
Recommended input voltage	200–240 Vac, 50/60 Hz
Low-line input support	100–127 Vac supported, but PSU output derates to 1000 W
High-line PSU output rating	1200 W per redundant PSU at 200–240 Vac
Estimated active DC load	~701 W DC
Estimated active AC input power	~730 W AC
Estimated high-load DC load	~834 W DC
Estimated high-load AC input power	~887 W AC
Recommended facility power allocation	**1.1 kW AC per server**
Nameplate allocation option	1.2 kW output capacity per server at 200–240 Vac
Estimated active current at 208 Vac	~3.7 A
Estimated active current at 240 Vac	~3.2 A
Estimated high-load current at 208 Vac	~4.5 A
Estimated high-load current at 240 Vac	~3.9 A
Recommended allocation current at 208 Vac	~5.6 A, based on 1.1 kW and 0.95 PF

Recommended allocation current at 240 Vac	~4.8 A, based on 1.1 kW and 0.95 PF
Estimated active heat output	~2,491 BTU/hr
Estimated high-load heat output	~3,026 BTU/hr
Recommended cooling allowance	~3,753 BTU/hr per server, based on 1.1 kW allocation

Component Power Basis

Component	Quantity	Power Basis	Subtotal
Intel Xeon Silver 4516Y+ CPU	2	185 W TDP each	370 W
NVIDIA ConnectX-7 single-port 400 GbE adapter	2	24.9 W typical each with passive cables	49.8 W
Micron 5400 MTFDDAK240TGA 240 GB SATA SSD	1	3.0 W sequential write typical	3 W
DDR5 ECC memory	1024 GB	Estimated, assuming 16 x 64 GB RDIMMs	~128 W active / ~160 W high-load
Motherboard, BMC, chipset, backplane, fans, cabling, platform overhead	1 system	Estimated	~150 W active / ~250 W high-load

Calculated active load:
370 W + 49.8 W + 3 W + 128 W + 150 W = 700.8 W DC

Estimated active AC input:
700.8 W ÷ 0.96 = ~730 W AC

High-load planning load:
370 W CPUs + 49.8 W NICs + 4 W SSD allowance + 160 W memory allowance + 250 W platform/fan allowance = ~834 W DC

Estimated high-load AC input:
834 W ÷ 0.94 = ~887 W AC

Supermicro lists the SYS-621C-TN12R with dual LGA-4677 sockets, 16 DDR5 DIMM slots, 12 hot-swap hybrid drive bays, up to 3 heavy-duty 8 cm fans, and **2 x 1200 W redundant Titanium Level 96% hot-plug power supplies**. The listed PSU output is **1000 W at 100–127 Vac** and **1200 W at 200–240 Vac**, so **200–240 Vac** is the preferred specification voltage for full PSU rating and rack-level margin. (Supermicro)

Intel lists the **Xeon Silver 4516Y+** as a 24-core processor with **48 threads** and **185 W TDP**; with two processors installed, the CPU TDP basis is **370 W**. (Intel)

For the boot SSD, Micron's 5400-series technical specification lists the **240 GB 2.5-inch SATA** power values as **1.5 W idle**, **3.0 W sequential write**, and **2.5 W sequential read**. I used **3.0 W** as the active SSD planning value. (Micron Assets)

For the network adapters, the exact card OPN was not provided beyond ConnectX-7 EN / MT2910, so I used NVIDIA's published single-port QSFP112 400 GbE-class ConnectX-7 reference, **MCX715105AS-WEAT**, as the closest power basis. NVIDIA lists this adapter as PCIe Gen5 x16, single QSFP112, 400/200/100/50/25/10 GbE-capable, Secure Boot enabled, Crypto disabled, with **24.9 W typical power using passive cables**. (NVIDIA Networking)

Specification-sheet note: The NIC estimate excludes QSFP112 optical transceiver or active cable power. Add the module power separately when optics are installed. The memory power is also an estimate because the DIMM count and exact DIMM model were not specified; the calculation assumes a 16-DIMM, 64 GB-per-DIMM configuration.

Data network switches

2 x NVIDIA SN5600 spine data network switches 6 x NVIDIA SN5600 leaf data network switches https://docs.nvidia.com/networking/display/sn5000/specifications#src-2705811927_Specifications-SN5600Specifications

Switches - Physical

- 2 x NVIDIA SN5600 spine data network switches
  - 800Gb Ethernet Switch (RDMA-capable)
  - 128 Total Port Count
  - 113 Used Port Count
  - These switches support full-speed RoCEv2 transport, and were configured with BGP (Border Gateway Protocol).
- 6 x NVIDIA SN5600 leaf data network switches
  - 800Gb Ethernet Switch (RDMA-capable)
  - 384 Total Port Count
  - 262 Used Port Count
  - These switches support full-speed RoCEv2 transport, and were configured with BGP (Border Gateway Protocol).

Data Network Switch Power Specifications

Switch platform: NVIDIA SN5600 Spectrum-4 Ethernet switch
Switch role: Spine and leaf data network switches
Network mode: 800 GbE Ethernet, RDMA/RoCEv2-capable, BGP-configured
Switch quantity: 8 total — 2 spine, 6 leaf

NVIDIA lists the SN5600 as a 2U Spectrum-4 switch with **64 OSFP data ports supporting 10/25/50/100/200/400/800G**, one SFP28 management/service port supporting **1/10/25G**, **51.2 Tbps throughput**, and **940 W typical power with passive cables**. The system uses **2 PSUs** and **4 fan modules**. (NVIDIA Docs)

Single-Switch Power Specifications

Specification	Value
System	NVIDIA SN5600
Form factor	2U, 19-inch rack mount
Switch ASIC	NVIDIA Spectrum-4
Data ports	64 x OSFP, 10/25/50/100/200/400/800G
Management/service port	1 x SFP28, 1/10/25G
Switch throughput	51.2 Tbps
Power supply configuration	2 x redundant AC PSUs
Fan configuration	4 fan modules, 3+1 fan redundancy
Recommended input voltage	200–264 Vac
Rated high-line input voltage	220–240 Vac, 50/60 Hz
Manufacturer AC current rating	16 A
Typical switch power, passive-cable baseline	**940 W AC per switch**
Recommended facility power allocation, passive-cable baseline	**1.2 kW AC per switch**
Estimated typical current at 208 Vac	~4.8 A, based on 940 W and 0.95 PF

Estimated typical current at 240 Vac	~4.1 A, based on 940 W and 0.95 PF
Recommended allocation current at 208 Vac	~6.1 A, based on 1.2 kW and 0.95 PF
Recommended allocation current at 240 Vac	~5.3 A, based on 1.2 kW and 0.95 PF
Estimated typical heat output	~3,207 BTU/hr per switch
Recommended cooling allowance	~4,095 BTU/hr per switch, based on 1.2 kW allocation

Switch Deployment Power Summary — Passive-Cable Baseline

Switch Group	Quantity	Total Port Count	Used Port Count	Typical Power	Recommended Facility Allocation	Typical Heat Output	Cooling Allowance
SN5600 spine switches	2	128	113	**1.88 kW**	**2.4 kW**	~6,415 BTU/hr	~8,189 BTU/hr
SN5600 leaf switches	6	384	262	**5.64 kW**	**7.2 kW**	~19,245 BTU/hr	~24,567 BTU/hr
Total data network switches	8	512	375	**7.52 kW**	**9.6 kW**	~25,659 BTU/hr	~32,757 BTU/hr

Optional Optical-Module Add-On

The **940 W per-switch** value is NVIDIA's typical SN5600 power figure **with passive cables**. Optical transceivers or active cables should be added separately. NVIDIA's published 800G OSFP transceiver examples for Spectrum-4-class switch links list **17 W maximum power per OSFP module**, so the following estimate uses **17 W per used port** as an optics-inclusive planning example. Use the actual module datasheet value when the exact optic or cable part number is known. (NVIDIA Networking)

Switch Group	Used Ports	Optical Add-On at 17 W/Port	Passive Baseline	Optics-Inclusive Estimate	Optics-Inclusive + 20% Headroom
SN5600 spine switches	113	1.92 kW	1.88 kW	**3.80 kW**	**4.56 kW**
SN5600 leaf switches	262	4.45 kW	5.64 kW	**10.09 kW**	**12.11 kW**
Total data network switches	375	6.38 kW	7.52 kW	**13.90 kW**	**16.67 kW**

Calculation Basis

Single-switch passive-cable baseline:
940 W AC

Recommended per-switch allocation:
940 W × 1.20 = 1,128 W, rounded to **1.2 kW AC per switch**

Single-switch typical current at 208 Vac:
940 W ÷ (208 V × 0.95 PF) = ~4.8 A

Single-switch typical current at 240 Vac:
940 W ÷ (240 V × 0.95 PF) = ~4.1 A

Single-switch heat output:
940 W × 3.412 = ~3,207 BTU/hr

Optional optics add-on:
Used ports × optical module power

For example, using 17 W per used OSFP port:

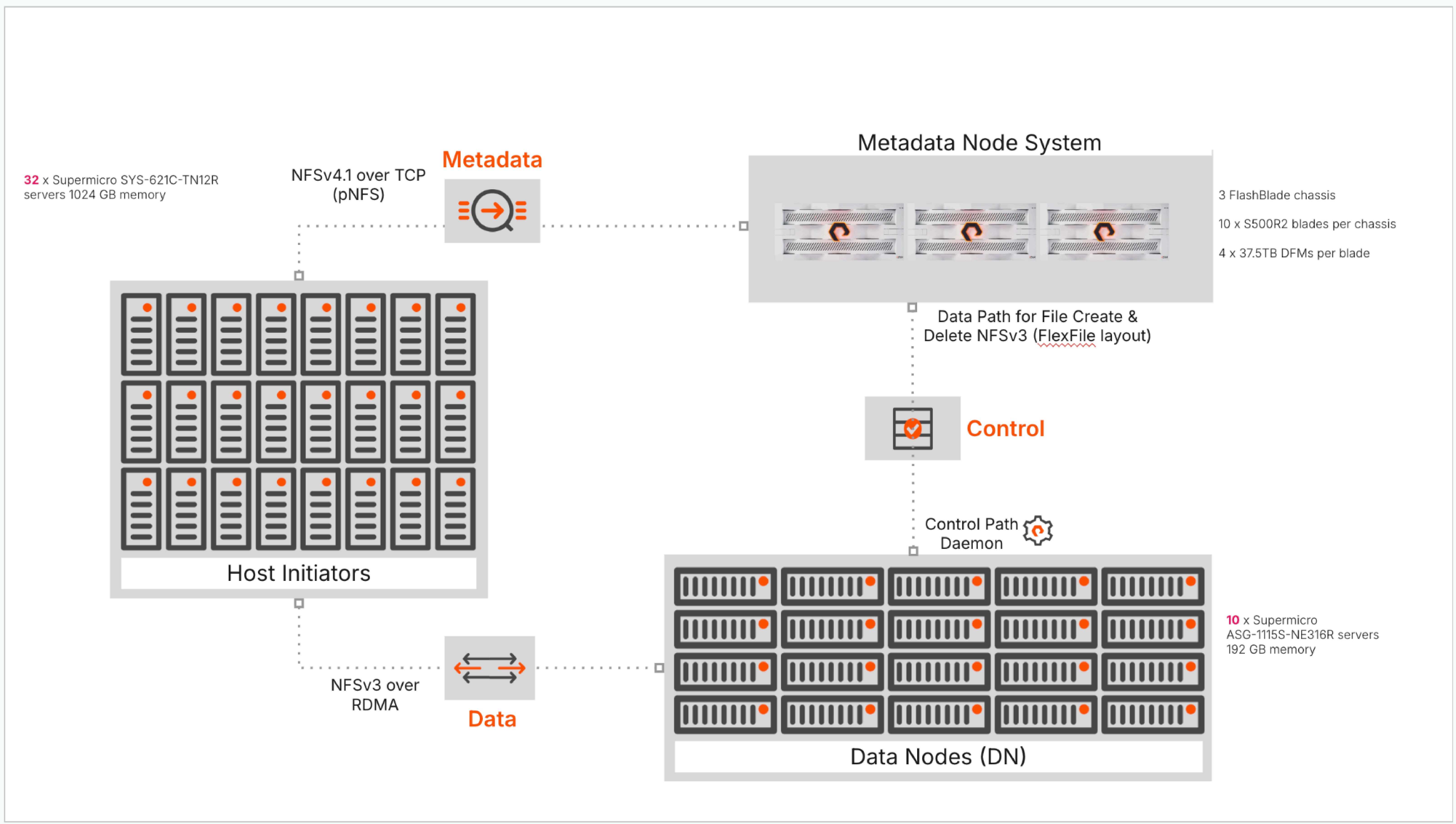
375 used ports × 17 W = 6,375 W

Optics-inclusive total estimate:
7.52 kW passive baseline + 6.38 kW optics add-on = ~13.90 kW

Specification-Sheet Note

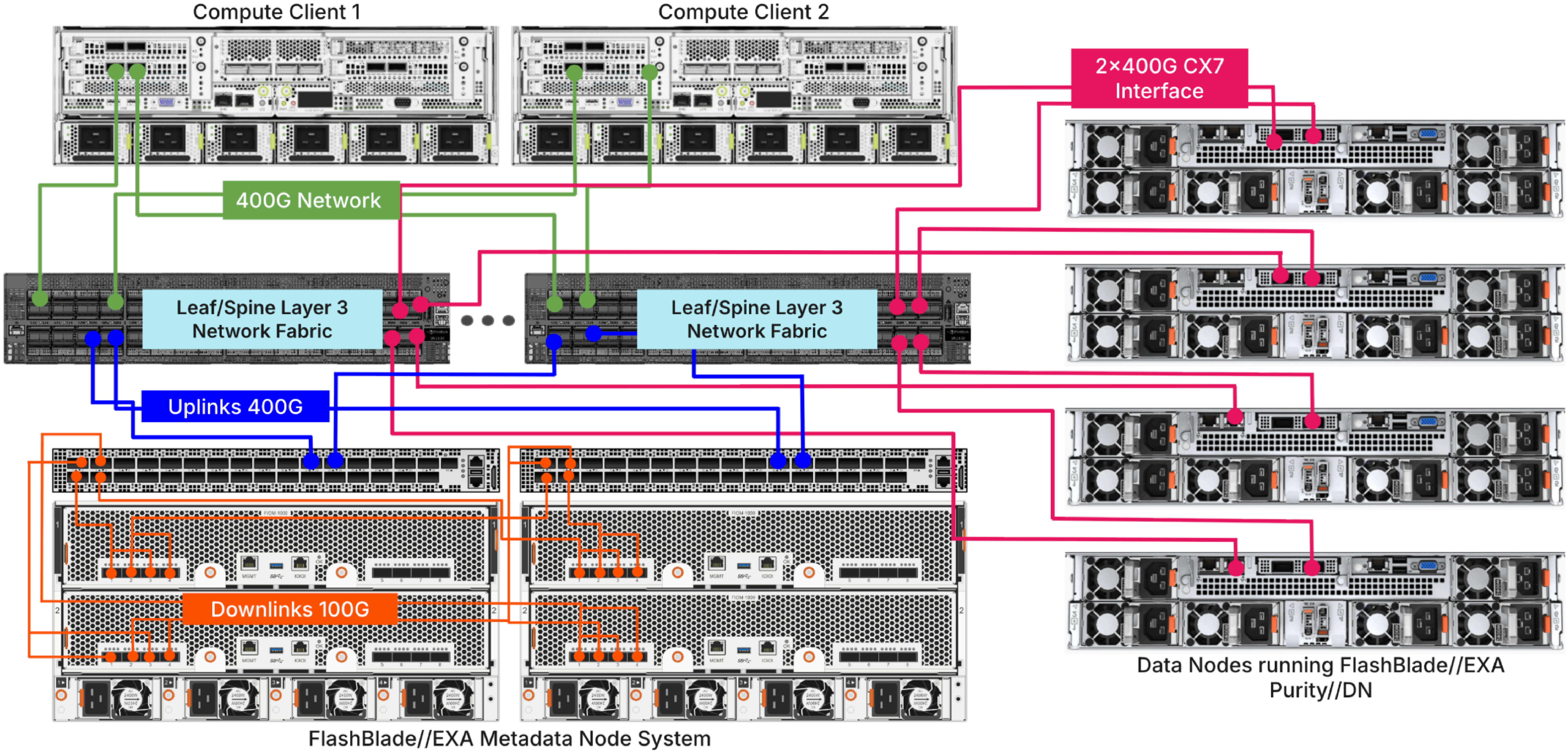
Specify the SN5600 switch fabric at **940 W AC typical per switch with passive cables**, and allocate **1.2 kW AC per switch** for facility planning when optics are not included. If optical OSFP modules are installed, add optical-module power separately by port count. With the provided used-port count and a 17 W/module planning value, the 8-switch data network estimate increases from **7.52 kW** to approximately **13.90 kW**, or **16.67 kW with 20% headroom**.

Pure Storage FlashBlade//EXA testbed logical diagram



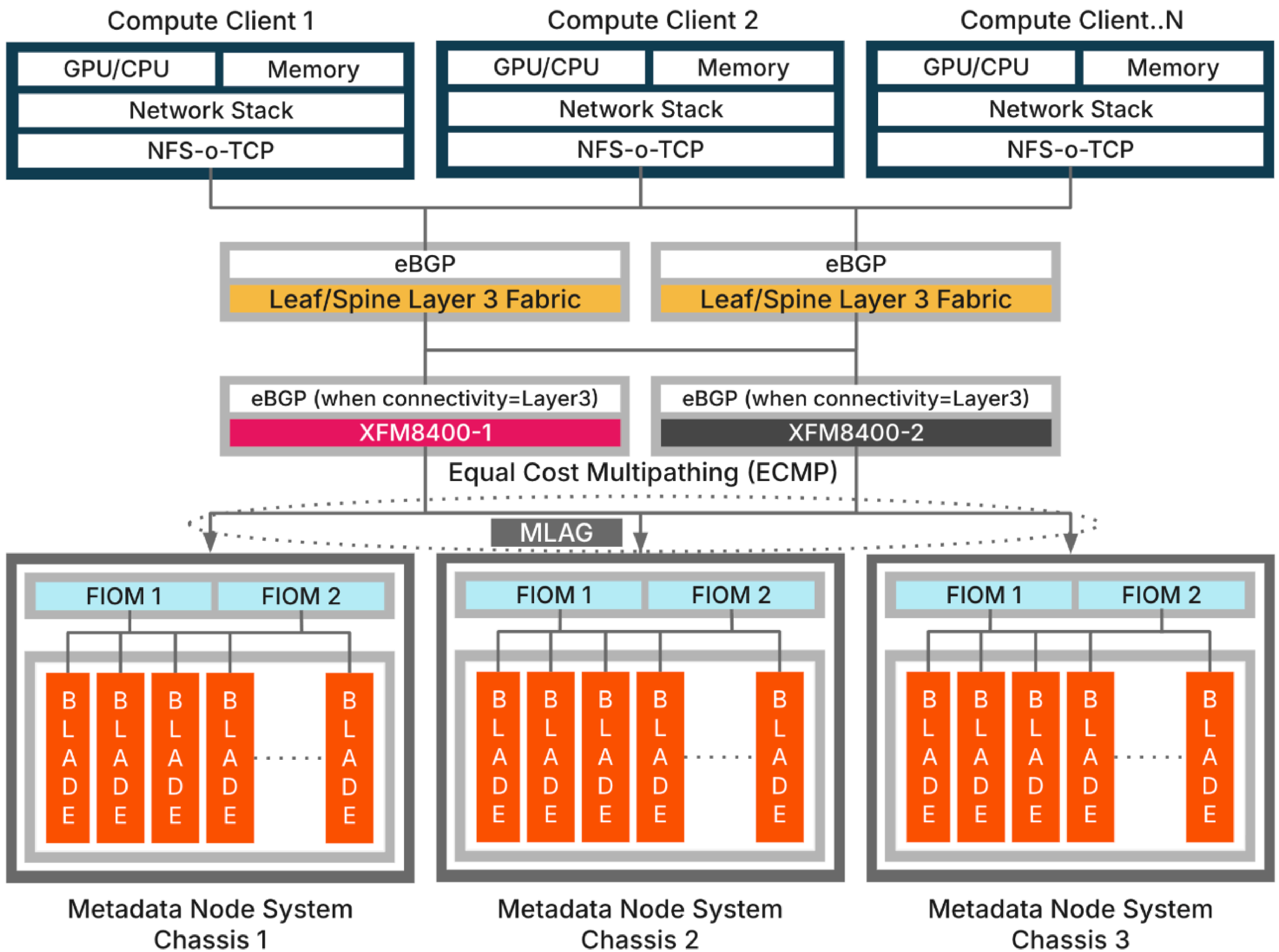
Pure Storage FlashBlade//EXA Physical Architecture diagram

Physical Architecture



Pure Storage FlashBlade//EXA Metadata Node Connectivity Stack diagram

Metadata Node Connectivity Stack



Metadata IO Stack

Compute Clients

Primary supercomputing clustered systems
Distributed parallel computing applications
Mix of GPU/CPU enabled data processing
pNFS over TCP network stack with eBGP

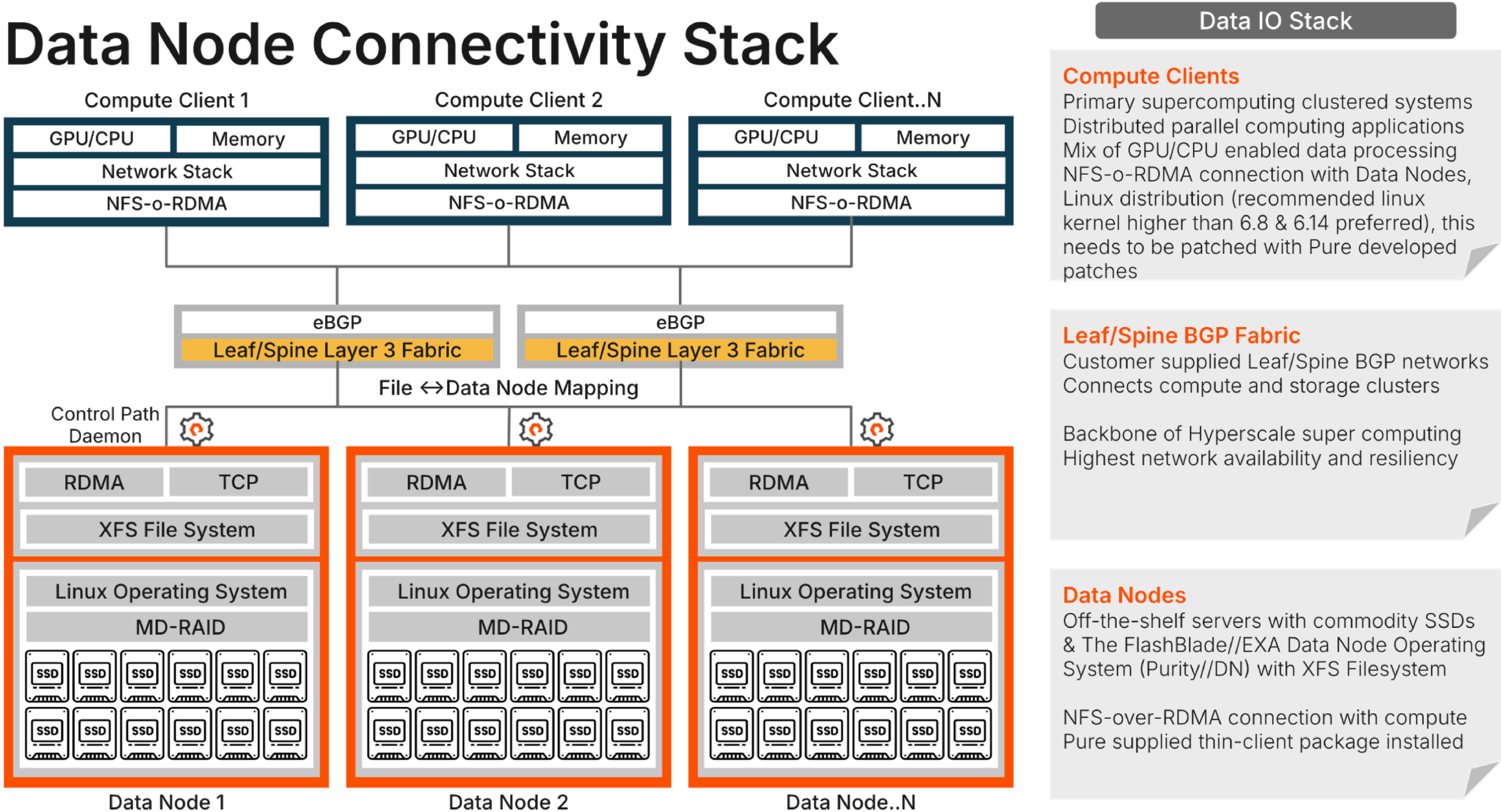
Leaf/Spine BGP Fabric

Customer supplied Leaf/Spine BGP networks
Connects compute and storage clusters
Backbone of Hyperscale super computing
Highest network availability and resiliency

Metadata Nodes

S500 Chassis, Purity OE & FIOMs are MLAG
Every XFM port has downlink (4×100G) to FIOM ports
XFM uplinks (4×400G) to Leaf/Spine switches
XFM complements eBGP network topology

Pure Storage FlashBlade//EXA Data Node Connectivity Stack diagram



Filesystem Creation Notes

The distributed filesystem was created via the FlashBlade//EXA GUI. It was configured with a NFSv4.1 export.

Storage and Filesystem Notes
When a filesystem is created on FlashBlade//EXA, the system orchestrates a series of coordinated actions across metadata node system (MDN) and data nodes (DNs), with a strong focus on scalability, performance, and data placement control.

1. Node Group Selection and Association Node Groups: Before a filesystem can be created, at least one data node group must exist. A node group is a logical collection of data nodes that will serve as the backing storage for the filesystem. This design allows administrators to control which DNs are used for specific filesystems, limiting the blast radius of failures and optimizing performance for different workloads. Association: When creating a filesystem, you must specify the node group it will use. All data for that filesystem will be placed only on the DNs in the selected group. This ensures that filesystems do not compete for IO resources across all DNs and that access can be maintained for unaffected files if a DN goes offline.

2. Filesystem Creation Workflow Metadata Node (MDN) Actions: The MDN receives the filesystem creation request (typically via the management GUI or CLI). It records the association between the new filesystem and the chosen node group. The MDN persists all relevant metadata, including the filesystem's unique ID, node group membership, and configuration details, in its distributed metadata store (DFMs). Data Node (DN) Preparation: The MDN communicates with each DN in the node group to prepare them for the new filesystem. On each DN, an XFS file system is created atop a software RAID (MD-RAID) array, using the local SSDs. The XFS export is assigned a UUID, which is tracked by the MDN. Export and Mounting: The DNs export the new XFS filesystem over NFS (typically NFSv3 over RDMA for data, NFSv4.1 over TCP for metadata). The MDN keeps a record of each export's UUID and IP address, ensuring that if a DN is replaced or its network changes, the system can recognize and re-associate the export.

3. Data Placement and File Mapping Placement Algorithm: When files are created within the new filesystem, the MDN uses a data placement algorithm to select which DN in the node group will store each file. The selection is based on available capacity, ensuring balanced utilization. At GA, each file is mapped to a single DN—striping across DNs is not supported. Metadata and Data Coordination: The MDN manages all metadata (directory structure, file attributes, etc.), while the DNs handle the actual file data. The MDN provides clients with the necessary information (DN IP, file handle, etc.) to access data directly on the appropriate DN.

4. Protocols and Control Protocols: The system uses NFSv4.1 (pNFS) for client-to-MDN communication and NFSv3 (FlexFile layout, often over RDMA) for client-to-DN data transfer. The MDN also uses gRPC and NFSv3 to coordinate with DNs for export management and health monitoring.

5. Manageability and Limits Node Group Constraints: You cannot create a filesystem with an empty node group, nor can you remove a DN from a node group if it is still in use by a live filesystem.
